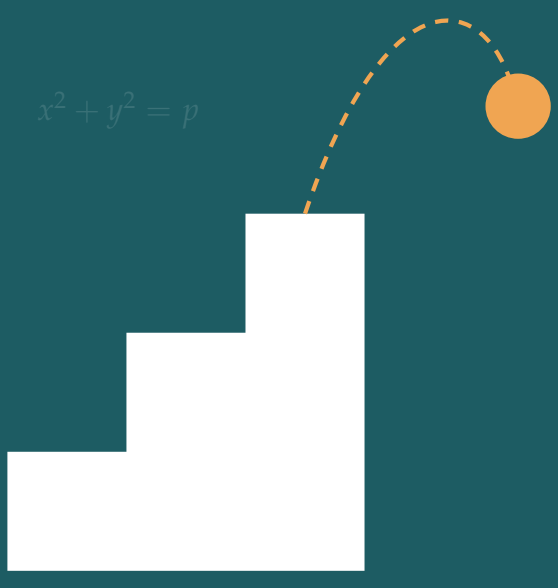
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP II AND III

2021 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP II AND III

2021 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

2021 Worked Solutions

Worked solutions to all 24 questions of the 2021 STEP papers (STEP II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

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WORKED SOLUTION

STEP II 2021, Question 1

Pure • Trigonometric Equations

STEP II 2021, Question 1

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

The key idea

Products of sines and cosines are awkward, but sums are easy to handle, so the opening identities let us trade a product for a sum. In (i) that turns a triple product into a cubic in $\cos 2x$ with a visible factor. In (ii) we clear the tangents, pair each factor with its partner of the opposite type, and the whole equation collapses into a product of two simple factors. The only extra care needed is over the points where a tangent is undefined.

Solution

The identities

Adding the expansions

$$\cos(A+B) = \cos A\cos B - \sin A\sin B, \qquad \cos(A-B) = \cos A\cos B + \sin A\sin B$$

gives $\cos(A+B)+\cos(A-B) = 2\cos A\cos B$. Taking $A = 3a$ and $B = a$,

$$\cos a\cos 3a = \tfrac{1}{2}(\cos 4a + \cos 2a),$$

as required. In the same way, adding $\sin(A \pm B) = \sin A\cos B \pm \cos A\sin B$ gives $\sin(A+B)+\sin(A-B) = 2\sin A\cos B$. With $A = a$ and $B = 3a$, and using $\sin(-2a) = -\sin 2a$,

$$\sin a\cos 3a = \tfrac{1}{2}(\sin 4a - \sin 2a).$$

(Swapping the roles, $A = 3a$ and $B = a$, gives the companion $\cos a\sin 3a = \tfrac{1}{2}(\sin 4a + \sin 2a)$, which we shall also need.)

Part (i)

By the first identity, $4\cos x\cos 2x\cos 3x = 2\cos 2x\,(\cos 4x + \cos 2x)$. Write $c = \cos 2x$, so that $\cos 4x = 2c^2 - 1$. The equation becomes

$$2c\,(2c^2 + c - 1) = 1 \quad\Longleftrightarrow\quad 4c^3 + 2c^2 - 2c - 1 = 0.$$

Grouping the terms in pairs, $2c^2(2c+1) - (2c+1) = 0$, that is,

$$(2c+1)(2c^2-1) = 0,$$

so $\cos 2x = -\frac{1}{2}$ or $\cos 2x = \pm\frac{1}{\sqrt{2}}$.

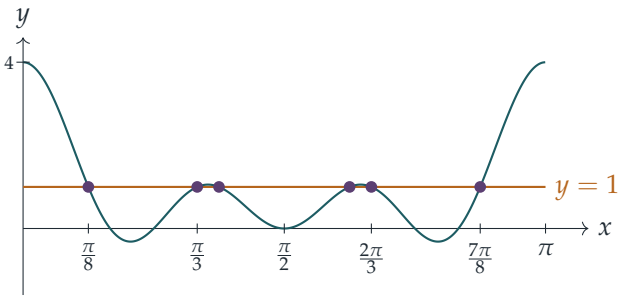
For $0 \le x \le \pi$ we have $0 \le 2x \le 2\pi$, and each value of $\cos 2x$ strictly between -1 and 1 gives two values of $2x$ in that range:

$$\begin{aligned}
\cos 2x = -\tfrac{1}{2} &\implies 2x = \tfrac{2\pi}{3},\ \tfrac{4\pi}{3} &&\implies x = \tfrac{\pi}{3},\ \tfrac{2\pi}{3};\\
\cos 2x = \tfrac{1}{\sqrt{2}} &\implies 2x = \tfrac{\pi}{4},\ \tfrac{7\pi}{4} &&\implies x = \tfrac{\pi}{8},\ \tfrac{7\pi}{8};\\
\cos 2x = -\tfrac{1}{\sqrt{2}} &\implies 2x = \tfrac{3\pi}{4},\ \tfrac{5\pi}{4} &&\implies x = \tfrac{3\pi}{8},\ \tfrac{5\pi}{8}.
\end{aligned}$$

Every step was reversible, so the solutions are

$$x = \frac{\pi}{8},\ \frac{\pi}{3},\ \frac{3\pi}{8},\ \frac{5\pi}{8},\ \frac{2\pi}{3},\ \frac{7\pi}{8}.$$

The graph confirms there are exactly six crossings of the line $y = 1$.



Part (ii)

Suppose $\tan x = \tan 2x \tan 3x \tan 4x$. For this even to make sense, all four tangents must be defined, so $\cos x$, $\cos 2x$, $\cos 3x$ and $\cos 4x$ are all non-zero. We may therefore multiply through by $\cos x \cos 2x \cos 3x \cos 4x$ to get

$$\sin x \cos 3x \cdot \cos 2x \cos 4x = \cos x \sin 3x \cdot \sin 2x \sin 4x.$$

? **Ask yourself** Why pair $\sin x$ with $\cos 3x$, and $\cos x$ with $\sin 3x$?

Because those are exactly the products our identities convert into sums, and both sums involve only $\sin 4x$ and $\sin 2x$, which already appear elsewhere in the equation. Using $\sin x \cos 3x = \frac{1}{2}(\sin 4x - \sin 2x)$ and $\cos x \sin 3x = \frac{1}{2}(\sin 4x + \sin 2x)$ and doubling,

$$(\sin 4x - \sin 2x)\cos 2x \cos 4x = (\sin 4x + \sin 2x)\sin 2x \sin 4x.$$

Now replace $\sin 2x \cos 2x$ by $\frac{1}{2}\sin 4x$ on the left, so that every term has a factor $\sin 4x$:

$$\sin 4x \cos 4x \cos 2x - \tfrac{1}{2}\sin 4x \cos 4x = \sin 4x \sin^2 2x + \sin 4x \sin 4x \sin 2x.$$

Taking everything to one side,

$$\sin 4x\left[(\cos 4x\cos 2x-\sin 4x\sin 2x)-\left(\sin^2 2x+\tfrac{1}{2}\cos 4x\right)\right]=0.$$

The first bracket is $\cos(4x+2x)=\cos 6x$. For the second, $\sin^2 2x=\frac{1}{2}(1-\cos 4x)$, so $\sin^2 2x+\frac{1}{2}\cos 4x=\frac{1}{2}$. Hence

$$\sin 4x\left(\cos 6x-\tfrac{1}{2}\right)=0,$$

so $\cos 6x=\frac{1}{2}$ or $\sin 4x=0$, as required.

Now for the solutions of $(\star)$ with $0\le x\le\pi$.

Ask yourself Is every solution of $\sin 4x(\cos 6x-\frac{1}{2})=0$ a solution of $(\star)$?

Not necessarily: the only step that could fail to reverse was multiplying by the cosines, and that is reversible precisely when all four tangents are defined. So the solutions of $(\star)$ are the solutions of $\sin 4x(\cos 6x-\frac{1}{2})=0$ at which $\cos x,\cos 2x,\cos 3x,\cos 4x\neq 0$.

Case $\sin 4x=0$. In $[0,\pi]$ this gives $x=0,\ \frac{\pi}{4},\ \frac{\pi}{2},\ \frac{3\pi}{4},\ \pi$. At $x=\frac{\pi}{4}$ and $x=\frac{3\pi}{4}$, $\tan 2x$ is undefined; at $x=\frac{\pi}{2}$, $\tan x$ is undefined. At $x=0$ and $x=\pi$ every tangent is 0, and $(\star)$ reads $0=0$. So $x=0$ and $x=\pi$ are solutions.

Case $\cos 6x=\frac{1}{2}$. Then $6x=\pm\frac{\pi}{3}+2k\pi$, so $x=\frac{(6k\pm1)\pi}{18}$. With $0\le 6x\le 6\pi$ this gives

$$x=\frac{\pi}{18},\ \frac{5\pi}{18},\ \frac{7\pi}{18},\ \frac{11\pi}{18},\ \frac{13\pi}{18},\ \frac{17\pi}{18}.$$

Each is $\frac{m\pi}{18}$ with m not a multiple of 3. A cosine $\cos nx$ vanishes only when nx is an odd multiple of $\frac{\pi}{2}=\frac{9\pi}{18}$, which would need $9\mid nm$. Since $3\nmid m$, this would need $9\mid n$, which is impossible for $n\in\{1,2,3,4\}$. So all four tangents are defined and these six values are all solutions.

Altogether $(\star)$ has eight solutions in $0\le x\le\pi$:

$$x=0,\ \frac{\pi}{18},\ \frac{5\pi}{18},\ \frac{7\pi}{18},\ \frac{11\pi}{18},\ \frac{13\pi}{18},\ \frac{17\pi}{18},\ \pi.$$

Checks

- In (i), at $x=\frac{\pi}{8}$: $\cos\frac{\pi}{4}=\frac{1}{\sqrt{2}}$ and $\cos\frac{3\pi}{8}\cos\frac{\pi}{8}=\frac{1}{2}(\cos\frac{\pi}{2}+\cos\frac{\pi}{4})=\frac{1}{2\sqrt{2}}$, so the product is $4\cdot\frac{1}{\sqrt{2}}\cdot\frac{1}{2\sqrt{2}}=1$.
- In (ii), at $x=\frac{\pi}{18}=10^\circ$: $\tan 10^\circ\approx 0.17633$, while $\tan 20^\circ\tan 30^\circ\tan 40^\circ\approx 0.36397\times 0.57735\times 0.83910\approx 0.17633$.
- A computer scan of both equations over $[0,\pi]$ finds exactly six and eight solutions respectively, at the values above.

Common slips

- Keeping $x = \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}$ in (ii): they satisfy $\sin 4x = 0$ but make a tangent in $(\star)$ undefined.
- Forgetting $x = 0$ and $x = \pi$, or missing the second value of $2x$ (or $6x$) in each period.
- Dividing by $\sin 4x$ during the simplification, which loses the factor that the question wants you to find.

Beyond STEP

The product-to-sum formulae are the real-variable shadow of $\mathrm{e}^{\mathrm{i}A}\mathrm{e}^{\mathrm{i}B} = \mathrm{e}^{\mathrm{i}(A+B)}$: writing $\cos\theta = \frac{1}{2}(\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta})$ turns any product of cosines into a sum, which is the first step towards Fourier series, where functions are expanded as sums of $\cos nx$ and $\sin nx$. The striking identity $\tan 10^\circ = \tan 20^\circ \tan 30^\circ \tan 40^\circ$ hidden in (ii) is typical of the surprises that appear when trigonometric values are viewed as roots of unity. Needham's *Visual Complex Analysis* develops this viewpoint beautifully.

WORKED SOLUTION

STEP II 2021, Question 2

Pure • Symmetric Functions of Roots

STEP II 2021, Question 2

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

The key idea

Sums of powers of two numbers are controlled by just two quantities: their sum and their product. Once you know $a+b$ and a^2+b^2, the product ab is forced, and so is a^3+b^3; that is the constraint in part (i). Part (iii) uses the same fact with c "removed": subtracting c, c^2, c^3 from the three-variable sums leaves two-variable sums, and the constraint from part (i) turns into a cubic equation that c must satisfy.

Solution

Part (i)

From $(a+b)^2 = a^2+2ab+b^2$ we get $p^2 = q+2ab$, so

$$ab = \frac{p^2-q}{2}.$$

Now use the factorisation $a^3+b^3 = (a+b)^3 - 3ab(a+b)$:

$$\begin{aligned} r = a^3+b^3 &= p^3 - 3p\cdot\frac{p^2-q}{2} \\ &= \frac{2p^3-3p^3+3pq}{2} = \frac{3pq-p^3}{2}. \end{aligned}$$

Hence $3pq-p^3 = 2r$, as required. Notice that nothing here used the fact that p, q, r are real: the identity holds for any complex a and b. We shall need that in part (iii).

Part (ii)

The quadratic $2x^2-2px+(p^2-q)=0$ always has two roots (possibly equal, possibly non-real), given by the quadratic formula. Call them a and b. Dividing through by 2, the quadratic is $x^2-px+\frac{1}{2}(p^2-q)$, so

$$a+b=p, \qquad ab=\frac{p^2-q}{2}.$$

This gives the first equation in $(*)$ directly. Then

$$a^2 + b^2 = (a+b)^2 - 2ab = p^2 - (p^2 - q) = q,$$

which is the second, and

$$a^3 + b^3 = (a+b)^3 - 3ab(a+b) = p^3 - \frac{3p(p^2-q)}{2} = \frac{3pq - p^3}{2} = r,$$

using the hypothesis $3pq - p^3 = 2r$ at the last step. So all three equations in $(*)$ hold.

The choice of quadratic was not luck: part (i) told us what ab has to be, and a quadratic with roots of known sum p and product $\frac{1}{2}(p^2 - q)$ is exactly $x^2 - px + \frac{1}{2}(p^2 - q)$.

Part (iii)

Set

$$p = a + b = s - c, \qquad q = a^2 + b^2 = t - c^2, \qquad r = a^3 + b^3 = u - c^3.$$

? **Ask yourself** Part (i) assumed p, q, r were real, but here they involve c, which may be complex. Can we still use it?

Yes: as noted at the end of part (i), the argument there is pure algebra and never used reality. So $3pq - p^3 = 2r$, that is,

$$3(s-c)(t-c^2) - (s-c)^3 = 2(u - c^3).$$

Expanding,

$$\begin{aligned} 3(s-c)(t-c^2) &= 3st - 3sc^2 - 3tc + 3c^3, \\ (s-c)^3 &= s^3 - 3s^2c + 3sc^2 - c^3, \end{aligned}$$

so the equation becomes

$$3st - 3sc^2 - 3tc + 3c^3 - s^3 + 3s^2c - 3sc^2 + c^3 = 2u - 2c^3.$$

Collecting powers of c:

$$6c^3 - 6sc^2 + 3(s^2 - t)c + 3st - s^3 - 2u = 0.$$

So c is a root of $6x^3 - 6sx^2 + 3(s^2 - t)x + 3st - s^3 - 2u = 0$, as required.

The coefficients depend only on s, t, u, which are symmetric in a, b, c. Repeating the argument with a (or b) in the role of c shows that a and b are roots too. So the other two roots are a and b.

? **Ask yourself** Why does it matter that a, b and c are distinct?

A cubic has at most three roots. Because a, b, c are three *different* roots, they are all of them, each simple, and so

$$6x^3 - 6sx^2 + 3(s^2 - t)x + 3st - s^3 - 2u = 6(x-a)(x-b)(x-c).$$

(If, say, $a = b$, we would only know two roots and could not identify the third.) Comparing constant terms, $3st - s^3 - 2u = -6abc = -6v$, that is,

$$s^3 - 3st + 2u = 6v.$$

Part (iv)

Here $s = 3$, $t = 1$, $u = -3$, $v = 2$. First a consistency check: the condition from part (iii) reads $27 - 9 - 6 = 12 = 6 \times 2$, so it holds. Part (iii) says that a, b, c should be the roots of

$$6x^3 - 18x^2 + 3(9-1)x + 9 - 27 + 6 = 6x^3 - 18x^2 + 24x - 12 = 0,$$

that is, of $x^3 - 3x^2 + 4x - 2 = 0$. Since $x = 1$ gives $1 - 3 + 4 - 2 = 0$,

$$x^3 - 3x^2 + 4x - 2 = (x-1)(x^2 - 2x + 2) = (x-1)\big((x-1)^2 + 1\big),$$

so the roots are 1, $1 + \mathrm{i}$, $1 - \mathrm{i}$. Take

$$a = 1, \qquad b = 1 + \mathrm{i}, \qquad c = 1 - \mathrm{i}.$$

Part (iii) only told us where to look, so we must now verify:

- $a + b + c = 1 + (1 + \mathrm{i}) + (1 - \mathrm{i}) = 3$;
- $(1 + \mathrm{i})^2 = 2\mathrm{i}$ and $(1 - \mathrm{i})^2 = -2\mathrm{i}$, so $a^2 + b^2 + c^2 = 1 + 2\mathrm{i} - 2\mathrm{i} = 1$;
- $(1 + \mathrm{i})^3 = 2\mathrm{i}(1 + \mathrm{i}) = -2 + 2\mathrm{i}$ and $(1 - \mathrm{i})^3 = -2\mathrm{i}(1 - \mathrm{i}) = -2 - 2\mathrm{i}$, so $a^3 + b^3 + c^3 = 1 - 4 = -3$;
- $abc = 1 \cdot (1 + \mathrm{i})(1 - \mathrm{i}) = 1 \cdot 2 = 2$.

All four equations $(**)$ hold. (Any rearrangement of these three numbers works equally well.)

Checks

- Part (i) with $a = 1$, $b = 2$: $p = 3$, $q = 5$, $r = 9$, and $3pq - p^3 = 45 - 27 = 18 = 2r$.
- Part (iii) with $a, b, c = 1, 2, 3$: $s = 6$, $t = 14$, $u = 36$, $v = 6$, and $s^3 - 3st + 2u = 216 - 252 + 72 = 36 = 6v$. The cubic is $6x^3 - 36x^2 + 66x - 36 = 6(x-1)(x-2)(x-3)$, as it should be.
- The identities in parts (i)–(iii) were also confirmed by computer for many random complex a, b, c.

Common slips

- In part (iii), saying "a and b are the other roots" without noting that the cubic's coefficients are symmetric in a, b, c.

- Deducing $s^3 - 3st + 2u = 6v$ from the product of roots without using distinctness to know that a, b, c are *all* the roots.
- In part (iv), stopping at the roots of the cubic: the question asks you to verify the four equations explicitly.

Beyond STEP

This question is a small case of Newton's identities, which link the power sums $a^k + b^k + c^k$ to the elementary symmetric functions $a + b + c$, $ab + bc + ca$, abc. The fundamental theorem of symmetric polynomials says that every symmetric polynomial can be written in terms of the elementary ones, and this is the starting point of Galois theory. For more, see Allenby, *Rings, Fields and Groups*.

WORKED SOLUTION

STEP II 2021, Question 3

Pure • Floor and Fractional Parts

STEP II 2021, Question 3

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

The key idea

Every real number splits uniquely into an integer part and a fractional part in $[0,1)$. So when an equation says "an integer plus a number in $[0,1)$ equals 4.9", both pieces are forced: the integer must be 4 and the fraction must be 0.9. In (ii) and (iii) the equations are tangled, but adding all three lets every $\lfloor t \rfloor + \{t\}$ recombine into t. That gives one clean linear equation, and subtracting each original equation from it separates the unknowns into exactly that "integer plus fraction" form.

Solution

Throughout we use: if n is an integer, $0 \leq f < 1$ and $n + f = c$, then $n = \lfloor c \rfloor$ and $f = \{c\}$. This holds because $n \leq c < n+1$, which pins down n as the largest integer not exceeding c.

Part (i)

In the first equation $\lfloor x \rfloor$ is an integer and $0 \leq \{y\} < 1$, so

$$\lfloor x \rfloor = 4, \qquad \{y\} = 0.9.$$

In the second, $\lfloor y \rfloor$ is an integer and $0 \leq \{x\} < 1$, so $\lfloor y \rfloor = \lfloor -1.4 \rfloor = -2$ and $\{x\} = -1.4 - (-2) = 0.6$. Hence

$$x = 4 + 0.6 = 4.6, \qquad y = -2 + 0.9 = -1.1.$$

These satisfy both equations ($4 + 0.9 = 4.9$ and $0.6 - 2 = -1.4$), and we have shown they are the only solution.

Part (ii)

Add the three equations. Each of x, y, z appears once on its own, and once more split as $\lfloor\cdot\rfloor + \{\cdot\}$ across two different equations:

$$(x+y+z) + (\{x\}+\lfloor x\rfloor) + (\lfloor y\rfloor + \{y\}) + (\{z\}+\lfloor z\rfloor) = 3.9 + 5.3 + 5,$$

that is, $2(x+y+z) = 14.2$, so

$$x+y+z = 7.1.$$

Ask yourself Now that we know $x+y+z$, what is left over when we take away one of the original equations?

Subtracting the first equation from $x+y+z = 7.1$ removes x and leaves $y - \lfloor y\rfloor$ and $z - \{z\}$:

$$\{y\} + \lfloor z\rfloor = 7.1 - 3.9 = 3.2,$$

which is the required result. In the same way, subtracting the second and third equations gives

$$\lfloor x\rfloor + \{z\} = 7.1 - 5.3 = 1.8, \qquad \{x\} + \lfloor y\rfloor = 7.1 - 5 = 2.1.$$

Each of these is "integer plus fractional part", so

$$\begin{aligned} \lfloor z\rfloor &= 3, & \{y\} &= 0.2,\\ \lfloor x\rfloor &= 1, & \{z\} &= 0.8,\\ \lfloor y\rfloor &= 2, & \{x\} &= 0.1. \end{aligned}$$

Therefore $x = 1.1$, $y = 2.2$, $z = 3.8$. This is the only possible solution, and it does work:

$$1.1 + 2 + 0.8 = 3.9, \qquad 0.1 + 2.2 + 3 = 5.3, \qquad 1 + 0.2 + 3.8 = 5.$$

Part (iii)

The same trick works. Adding the three equations, with $2\lfloor y\rfloor + 2\{y\} = 2y$,

$$2x + 4y + 2z = 14.2, \qquad \text{so} \qquad x + 2y + z = 7.1.$$

Subtracting each equation in turn from this gives

$$2\{y\} + \lfloor z\rfloor = 3.2, \qquad \lfloor x\rfloor + \{z\} = 1.8, \qquad \{x\} + 2\lfloor y\rfloor = 2.1.$$

The middle equation is of the familiar type, so $\lfloor x\rfloor = 1$ and $\{z\} = 0.8$.

In the third equation, $2\lfloor y\rfloor$ is still an integer and $0 \le \{x\} < 1$, so the earlier reasoning applies: $2\lfloor y\rfloor = 2$ and $\{x\} = 0.1$. Hence $\lfloor y\rfloor = 1$ and $x = 1.1$.

Ask yourself Does the first equation still force a unique answer?

No. Here $2\{y\}$ lies in $[0, 2)$, not $[0, 1)$, so there are two ways to write 3.2 as an integer plus a number in $[0, 2)$:

- $\lfloor z \rfloor = 3$, $2\{y\} = 0.2$, so $\{y\} = 0.1$;
- $\lfloor z \rfloor = 2$, $2\{y\} = 1.2$, so $\{y\} = 0.6$.

(If $\lfloor z \rfloor \geq 4$ then $2\{y\} < 0$, and if $\lfloor z \rfloor \leq 1$ then $2\{y\} \geq 2.2$; both are impossible.) With $\lfloor y \rfloor = 1$ and $\{z\} = 0.8$, this gives two candidates:

$$(x, y, z) = (1.1,\ 1.1,\ 3.8) \qquad \text{or} \qquad (x, y, z) = (1.1,\ 1.6,\ 2.8).$$

Both satisfy the original system:

$$\begin{aligned}(1.1, 1.1, 3.8): &\quad 1.1 + 2 + 0.8 = 3.9, \quad 0.1 + 2.2 + 3 = 5.3, \quad 1 + 0.2 + 3.8 = 5;\\ (1.1, 1.6, 2.8): &\quad 1.1 + 2 + 0.8 = 3.9, \quad 0.1 + 3.2 + 2 = 5.3, \quad 1 + 1.2 + 2.8 = 5.\end{aligned}$$

So the system in (iii) has exactly these two solutions.

Checks

Substitute the answers back in, as above: every equation balances. As an independent check, a computer search with x and y running over all multiples of 0.1 in $[-10, 10]$ found exactly one solution to (i), one to (ii) and exactly the two above for (iii). Note that $\{y\} + \lfloor z \rfloor = 0.2 + 3 = 3.2$ for the solution of (ii), as it should.

Common slips

- Getting the floor of a negative number wrong: $\lfloor -1.4 \rfloor = -2$, not -1, so $\{x\} = 0.6$ in (i).
- In (iii), assuming $2\{y\}$ is a fractional part and so finding only one solution. It lies in $[0, 2)$, which is why a second solution appears.
- Not checking the candidates in the original equations. The deductions above only show they are the only possibilities.

Beyond STEP

Writing $x = \lfloor x \rfloor + \{x\}$ is the same as working modulo 1: the fractional part is the image of x in the circle $\mathbb{R}/\mathbb{Z}$. Floor and fractional parts are everywhere in number theory, for example in Legendre's formula for the power of a prime dividing $n!$, and in the study of how the fractional parts of $n\alpha$ spread out over $[0, 1)$ for irrational α. Davenport's *The Higher Arithmetic* is a lovely introduction to this circle of ideas.

WORKED SOLUTION

STEP II 2021, Question 4

Pure • Lambert W Function

STEP II 2021, Question 4

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 3.9, *Inventing a New Function*: https://beyondstep.keithmcnulty.org/ch03/lambert-w.html

The key idea

The function g undoes xe^x, so the whole question is about rearranging each equation into the shape $u\,e^u = c$ for some new variable u. Once it is in that shape, the sketch from part (i) tells us everything: there is no solution if $c < -\frac{1}{e}$, and two solutions if $-\frac{1}{e} < c < 0$, of which g only ever picks out the one with $u \geq -1$. Logarithms are dealt with by substituting $x = e^u$.

Solution

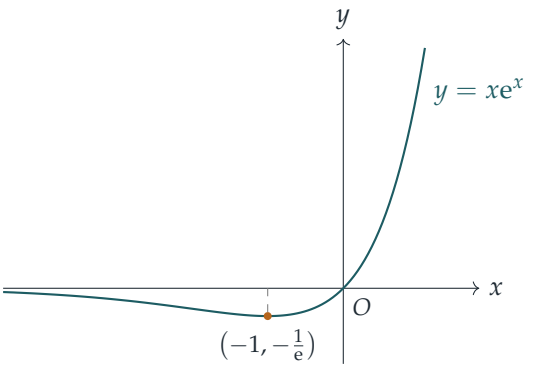
Part (i)

Let $y = xe^x$. By the product rule,

$$\frac{dy}{dx} = (1+x)e^x,$$

which is zero only at $x = -1$, since $e^x > 0$. The gradient is negative for $x < -1$ and positive for $x > -1$, so the only stationary point is a minimum at $\left(-1, -\frac{1}{e}\right)$.

The curve has the sign of x and passes through the origin. As $x \to \infty$, $y \to \infty$ rapidly. As $x \to -\infty$, e^x beats x, so $y \to 0$ from below: the negative x-axis is an asymptote.



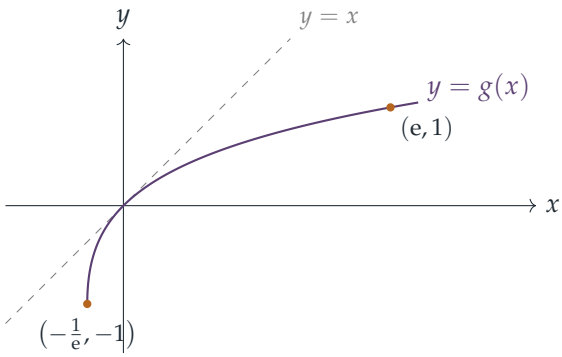
Part (ii)

For f to have an inverse it must be one-to-one on $[a, \infty)$. If $a < -1$, the interval contains points on both sides of the minimum near $x = -1$, and since f is continuous and strictly decreasing on $(-\infty, -1]$ and strictly increasing on $[-1, \infty)$, some value just above $-\frac{1}{\mathrm{e}}$ is taken twice. If $a \geq -1$, f is strictly increasing, hence one-to-one. So the minimum possible value is

$$a = -1.$$

Then f maps $[-1, \infty)$ onto $[-\frac{1}{\mathrm{e}}, \infty)$, so g has domain $[-\frac{1}{\mathrm{e}}, \infty)$ and range $[-1, \infty)$.

The graph of g is the reflection of the right-hand branch of part (i) in the line $y = x$. It starts at $(-\frac{1}{\mathrm{e}}, -1)$ with a vertical tangent (the reflection of the horizontal tangent at the minimum), passes through the origin, where it has gradient 1 (as $f'(0) = 1$), and increases ever more slowly: $g(\mathrm{e}) = 1$, and $g(x)$ grows more slowly than $\ln x$ for large x.



Two facts will be used repeatedly in part (iii). For $c \geq -\frac{1}{\mathrm{e}}$, $u = g(c)$ is the solution of $u\mathrm{e}^u = c$ with $u \geq -1$. And from the sketch in part (i), the equation $u\mathrm{e}^u = c$ has

- no real solution if $c < -\frac{1}{\mathrm{e}}$;
- two real solutions if $-\frac{1}{\mathrm{e}} < c < 0$, namely $g(c) \in (-1, 0)$ and another less than -1;
- exactly one real solution, $g(c)$, if $c \geq 0$ (for $u < 0$ the left-hand side is negative).

Part (iii)

(a) Multiplying by e^x (which is never zero) gives

$$xe^x = \tfrac{1}{5}.$$

Since $\frac{1}{5} > 0$, there is exactly one real root:

$$x = g(\tfrac{1}{5}).$$

(b) We need $x > 0$ for $\ln x$ to exist, so put $x = e^u$ with $u = \ln x$ real. The equation becomes

$$2ue^u + 1 = 0 \iff ue^u = -\tfrac{1}{2}.$$

But $ue^u \geq -\frac{1}{e}$ for all u, and $-\frac{1}{2} < -\frac{1}{e}$ because $e > 2$. So there is *no real root*.

(c) With $x = e^u$ as before,

$$3ue^u + 1 = 0 \iff ue^u = -\tfrac{1}{3}.$$

Now $-\frac{1}{e} < -\frac{1}{3} < 0$ because $e < 3$, so there are two values of u, and hence (as $u \mapsto e^u$ is one-to-one) two real roots x. One of them is

$$x = e^{g(-1/3)}.$$

? **Ask yourself** Is this the larger or the smaller root?

The other value of u is less than -1, while $g(-\frac{1}{3}) > -1$. Since e^u is increasing in u, the root $e^{g(-1/3)}$ is the *greater* of the two roots.

(d) Again $x > 0$. Exponentiating, $x = 3\ln x \iff e^{x/3} = x \iff xe^{-x/3} = 1$. To get the form ue^u we need the multiplier in front to match the exponent, so put $u = -\frac{x}{3}$ and multiply by $-\frac{1}{3}$:

$$\left(-\frac{x}{3}\right)e^{-x/3} = -\frac{1}{3}.$$

As in (c), $-\frac{1}{e} < -\frac{1}{3} < 0$, so there are two values of u, both negative, giving two positive roots $x = -3u$. One root is

$$x = -3\,g(-\tfrac{1}{3}).$$

This time the map $u \mapsto x = -3u$ is *decreasing*. The other value of u is less than -1, so the other root is greater than 3, while our root satisfies $x = -3g(-\frac{1}{3}) < 3$. So the root we have found is the *smaller* of the two.

Part (iv)

Since $x > 0$, take logarithms: $x^x = 10 \iff x\ln x = \ln 10$. Put $x = e^u$:

$$ue^u = \ln 10.$$

As $\ln 10 > 0$, this has the single solution $u = g(\ln 10)$, so the root is

$$x = e^{g(\ln 10)}.$$

(Equivalently, since $e^{g(c)} = c/g(c)$ for $c \neq 0$, the root is $\dfrac{\ln 10}{g(\ln 10)}$.)

Checks

- Numerically, $g(-\frac{1}{3}) \approx -0.6191$. In (c) this gives $x \approx 0.5384$, and $3(0.5384)\ln(0.5384) \approx -1.000$; the other root is about 0.2204, which is indeed smaller.
- In (d), $x \approx 1.8572$, and $3\ln 1.8572 \approx 1.8572$; the other root is about $4.536 > 3$, as predicted.
- In (iv), $g(\ln 10) \approx 0.9188$, so $x \approx 2.5062$, and $2.5062^{2.5062} \approx 10.00$.
- In (b), the smallest value of $2x\ln x + 1$ is $1 - \frac{2}{\mathrm{e}} \approx 0.264 > 0$ (at $x = \frac{1}{\mathrm{e}}$), confirming there is no root.

Common slips

- Taking $a = 0$ in part (ii) because the curve "looks" one-to-one there: the turning point is at $x = -1$, not at the origin.
- In (d), forgetting to multiply through by $-\frac{1}{3}$, so that the factor in front does not match the exponent and g cannot be applied.
- Assuming the root found via g is always the larger one: it is in (c), where $x = \mathrm{e}^u$ is increasing in u, but not in (d), where $x = -3u$ reverses the order.

Beyond STEP

The function g is the principal branch of the *Lambert W function*, usually written W_0; the other solution of $u\mathrm{e}^u = c$ for $-\frac{1}{\mathrm{e}} < c < 0$ comes from a second branch, W_{-1}. It turns up in delay differential equations, in combinatorics (counting labelled trees) and in physics, wherever an unknown appears both inside and outside an exponential. Continuing an inverse function across the point where it fails to be one-to-one is the start of the theory of branches and branch points in complex analysis; Needham's *Visual Complex Analysis* is a beautiful place to meet these ideas.

WORKED SOLUTION

STEP II 2021, Question 5

Pure • First-Order Differential Equations

STEP II 2021, Question 5

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

The key idea

Part (i) is a first-order equation that is neither separable nor obviously linear in a friendly form, but the suggested substitution turns it into one we can integrate directly. Part (ii) looks geometric, yet once we write down the equation of the tangent and insist that it passes through $(1, t)$, we get exactly the equation of part (i) with $a = 1$. The sketches then come from the derivative, which turns out to be remarkably simple.

Solution

Part (i)

The equation only makes sense for $x \neq a$, so we work on an interval where $x > a$ or one where $x < a$. With $y = (x-a)u$, the product rule gives

$$\frac{\mathrm{d}y}{\mathrm{d}x} = u + (x-a)\frac{\mathrm{d}u}{\mathrm{d}x}.$$

Substituting into the equation,

$$(x-a)u + (x-a)^2\frac{\mathrm{d}u}{\mathrm{d}x} = (x-a)u - x \quad \Longrightarrow \quad \frac{\mathrm{d}u}{\mathrm{d}x} = -\frac{x}{(x-a)^2}.$$

The u terms cancel, which is the whole point of the substitution. Writing $x = (x-a) + a$,

$$\frac{\mathrm{d}u}{\mathrm{d}x} = -\frac{1}{x-a} - \frac{a}{(x-a)^2}, \qquad \text{so} \qquad u = -\ln|x-a| + \frac{a}{x-a} + c$$

for a constant c. Multiplying by $x - a$,

$$\boxed{y = a + (x-a)\big(c - \ln|x-a|\big)}$$

where the constant c may differ on the two sides of $x = a$.

To check: $y' = c - \ln|x-a| - 1$, so $(x-a)y' = (y-a) - (x-a) = y - x$, as required.

Part (ii)

The tangent at $(t, \mathrm{f}(t))$ is $y = \mathrm{f}(t) + \mathrm{f}'(t)(x - t)$. It passes through $(1, t)$ exactly when

$$t = \mathrm{f}(t) + \mathrm{f}'(t)(1 - t), \qquad \text{that is,} \qquad (t - 1)\,\mathrm{f}'(t) = \mathrm{f}(t) - t.$$

Renaming t as x, this is the equation of part (i) with $a = 1$, holding for $x \neq 1$. Hence on each of $x < 1$ and $x > 1$,

$$\mathrm{f}(x) = 1 + (x - 1)\big(c - \ln|x - 1|\big),$$

with a constant that we fix separately on each side.

? **Ask yourself** Why can't one condition, such as $\mathrm{f}(0) = 0$, fix the curve for all x?

The equation tells us nothing at $x = 1$, so the solution on $x < 1$ and the solution on $x > 1$ are independent. That is why the question gives a separate condition for each part.

(a) For $x < 1$, $|x - 1| = 1 - x$. The condition $\mathrm{f}(0) = 0$ gives $0 = 1 + (-1)(c - \ln 1) = 1 - c$, so $c = 1$ and

$$\boxed{\mathrm{f}(x) = 1 + (x - 1)\big(1 - \ln(1 - x)\big) \quad (x < 1).}$$

Differentiating, $\mathrm{f}'(x) = 1 - \ln(1 - x) + (x - 1) \cdot \dfrac{1}{1 - x} = -\ln(1 - x)$. So:

- $\mathrm{f}'(x) = 0$ only when $1 - x = 1$, giving the single stationary point $(0, 0)$. Since $\mathrm{f}''(x) = \dfrac{1}{1 - x} > 0$ for $x < 1$, the curve is convex and $(0, 0)$ is a minimum. So $\mathrm{f}(x) \geq 0$, and the curve just touches the x-axis at the origin.
- As $x \to 1^-$, put $z = x - 1 \to 0$: $\mathrm{f}(x) = 1 + z - z \ln|z| \to 1$, using $z \ln|z| \to 0$. The gradient $-\ln(1 - x) \to +\infty$, so the curve heads towards $(1, 1)$ and becomes vertical there.
- As $x \to -\infty$, both $x - 1$ and $1 - \ln(1 - x)$ are large and negative, so $\mathrm{f}(x) \to +\infty$.

Some helpful points: $\mathrm{f}(1 - \mathrm{e}) = 1$ and $\mathrm{f}(-1) = 2 \ln 2 - 1 \approx 0.39$.

(b) For $x > 1$, $|x - 1| = x - 1$. The condition $\mathrm{f}(2) = 2$ gives $2 = 1 + 1 \cdot (c - 0)$, so again $c = 1$ and

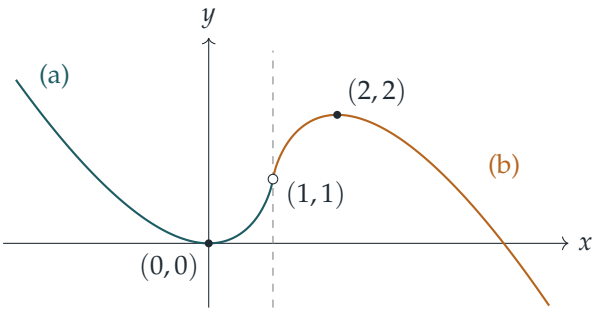
$$\mathrm{f}(x) = 1 + (x - 1)\big(1 - \ln(x - 1)\big) \quad (x > 1), \qquad \mathrm{f}'(x) = -\ln(x - 1).$$

- The only stationary point is at $x = 2$, namely $(2, 2)$. Here $\mathrm{f}''(x) = -\dfrac{1}{x - 1} < 0$, so the curve is concave and $(2, 2)$ is a maximum.
- As $x \to 1^+$, $\mathrm{f}(x) \to 1$ (by the same limit) and $\mathrm{f}'(x) \to +\infty$: the curve leaves $(1, 1)$ vertically.

- As $x \to \infty$, $1 - \ln(x-1) \to -\infty$, so $f(x) \to -\infty$. The curve passes through $(1+e, 1)$ and crosses the x-axis once, near $x \approx 4.59$.

Ask yourself Is there a quicker way to see the shape of (b) once (a) is done?

Yes. Writing $s = |x-1|$, the branch in (a) is $f(1-s) = 1 - s(1 - \ln s)$, and the branch in (b) is $f(1+s) = 1 + s(1 - \ln s)$. So $f(1+s) - 1 = -(f(1-s) - 1)$: the branch in (b) is the branch in (a) rotated through a half-turn about $(1,1)$. The minimum $(0,0)$ maps to the maximum $(2,2)$.



The curve is not defined at $x = 1$ (the open circle), but both branches approach $(1,1)$ with vertical tangents.

Checks

- Test the tangent property directly. At $t = 0$ the tangent is horizontal, $y = 0$, and it does pass through $(1, 0)$. At $t = 2$ the tangent is $y = 2$, which passes through $(1, 2)$.
- At $t = -1$: $f(-1) \approx 0.386$ and $f'(-1) = -\ln 2 \approx -0.693$. The tangent at $x = 1$ has height $0.386 + 2(-0.693) \approx -1.000$, which equals t.
- The symmetry check: $f(1+s) + f(1-s) = 2$ for every $s > 0$, for example $f(4) + f(-2) = (4 - 3\ln 3) + (3\ln 3 - 2) = 2$.

Common slips

- Forgetting the modulus in $\ln|x-a|$, and then being unable to handle $x < 1$ in part (a).
- Assuming that the constant found in (a) automatically applies for $x > 1$. Here it happens to come out the same, but that has to be checked using $f(2) = 2$.
- Concluding that $f(x) \to 0$ or blows up as $x \to 1$. The term $(x-1)\ln|x-1|$ tends to 0, so $f \to 1$. It is the gradient that blows up.

Beyond STEP

The equation $(x-a)y' = y-x$ is *homogeneous* once we shift the origin to $x=a$ and $y=a$: with $X = x-a$ and $Y = y-a$ it becomes $XY' = Y-X$. The substitution $Y = Xu$ is then the standard method for homogeneous equations. The point $x=a$, where the coefficient of y' vanishes, is a *singular point*, and the solutions on the two sides of a singular point need not be related at all. Boyce and DiPrima's *Elementary Differential Equations* treats both ideas.

WORKED SOLUTION

STEP II 2021, Question 6

Pure • Circle Geometry Approximations

STEP II 2021, Question 6

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

The key idea

Everything happens in one triangle: A, the centre O of the road, and the centre O' of the cyclist's circle. Symmetry puts O' on the line through O that bisects $\angle AOB$, and the tangency fixes the distance OO'. The cosine rule and the sine rule in that triangle then give d and $\alpha - \theta$. For the last part, *much less than 1* means we keep first-order terms in w/R and use small-angle approximations.

Solution

Take O as origin with the line of symmetry vertical, so that $A = \big(-(R+w)\sin\alpha,\ (R+w)\cos\alpha\big)$ and B is its mirror image. The road is the region $R \le r \le R+w$, where r is the distance from O.

Part (i)

The point of the chord AB closest to O is its midpoint M, since $OM \perp AB$ (triangle OAB is isosceles). Then

$$OM = (R+w)\cos\alpha \le \tfrac{1}{2}(R+w) < \tfrac{1}{2}(R+R) = R,$$

using $\cos\alpha \le \frac{1}{2}$ (because $\frac{1}{3}\pi \le \alpha \le \frac{1}{2}\pi$) and then $w < R$. So M lies strictly inside the inner circle, which is off the road. Any straight ride from A to B passes through M, so it cannot stay on the road.

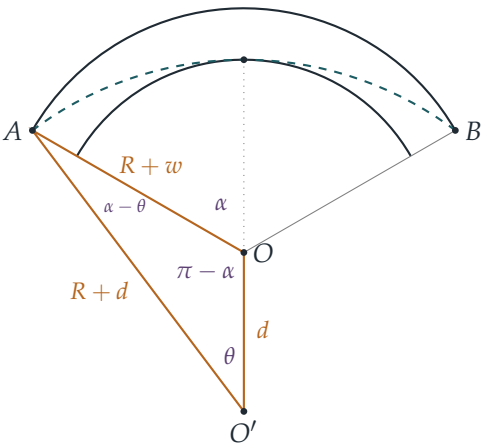
Part (ii)

Where is O'? The circle passes through A and B, so its centre is equidistant from them and lies on the perpendicular bisector of AB, the line of symmetry through O. It touches the inner circle and is bigger than it ($R + d > R$), so the inner circle sits inside it, touching it internally. For internally tangent circles the distance between centres is

the difference of the radii:

$$OO' = (R + d) - R = d.$$

The path bulges outwards, towards the road, so O' lies on the opposite side of O from the road, as in the figure. In triangle AOO' we then have $OA = R + w$, $OO' = d$, $O'A = R + d$, and the angle at O is $\pi - \alpha$ (it is supplementary to $\angle AOM = \alpha$). The angle at O' is $\frac{1}{2}\angle AO'B = \theta$, so the angle at A is $\pi - (\pi - \alpha) - \theta = \alpha - \theta$.



(a) The cosine rule in triangle AOO', with $\cos(\pi - \alpha) = -\cos\alpha$, gives

$$\begin{aligned}(R + d)^2 &= (R + w)^2 + d^2 + 2(R + w)d\cos\alpha \\ R^2 + 2Rd &= R^2 + 2Rw + w^2 + 2(R + w)d\cos\alpha \\ 2d\big(R - (R + w)\cos\alpha\big) &= w(2R + w),\end{aligned}$$

so

$$d = \frac{w(2R + w)}{2\big(R - (R + w)\cos\alpha\big)}.$$

The bracket in the denominator is positive by part (i), so this is a sensible positive length.

(b) The sine rule in the same triangle, pairing the angle $\alpha - \theta$ (opposite $OO' = d$) with the angle $\pi - \alpha$ (opposite $O'A = R + d$), gives

$$\frac{\sin(\alpha - \theta)}{d} = \frac{\sin(\pi - \alpha)}{R + d} \quad \Longrightarrow \quad \sin(\alpha - \theta) = \frac{d\sin\alpha}{R + d}.$$

Part (iii)

Write $\varepsilon = w/R$, so $0 < \varepsilon \ll 1$. Dividing the result of (a) by R,

$$\frac{d}{R} = \frac{\varepsilon(2 + \varepsilon)}{2\big(1 - (1 + \varepsilon)\cos\alpha\big)}.$$

Since $\cos\alpha \le \frac{1}{2}$, the denominator is at least $2\big(1-\frac{1}{2}(1+\varepsilon)\big) = 1-\varepsilon$, so

$$0 < \frac{d}{R} \le \frac{\varepsilon(2+\varepsilon)}{1-\varepsilon} \approx 2\varepsilon,$$

which is much less than 1. (To first order, $d/R \approx \varepsilon/(1-\cos\alpha)$.)

Ask yourself A small sine only tells us that the angle is close to 0 *or* close to π. Which is it?

From (b), $0 < \sin(\alpha-\theta) = \dfrac{(d/R)\sin\alpha}{1+d/R} < \dfrac{d}{R}$, which is small. The angle $\alpha-\theta$ is an angle of triangle AOO', so it is positive, and it is less than $\alpha \le \frac{1}{2}\pi$ because $\theta > 0$. On $(0, \frac{1}{2}\pi)$ the sine is increasing with $\sin x \ge \frac{2}{\pi}x$, so $\alpha-\theta \le \frac{\pi}{2}\sin(\alpha-\theta) < \frac{\pi}{2}\cdot\frac{d}{R}$, which is also much less than 1.

Part (iv)

My friend rides along the outer edge, an arc of radius $R+w$ subtending 2α at O, so their path has length $L = 2\alpha(R+w)$. My path is the arc of radius $R+d$ on the far side of AB from O'; it subtends 2θ at O', so its length is $2\theta(R+d)$. Therefore

$$\begin{aligned} S &= 2\alpha(R+w) - 2\theta(R+d) \\ &= 2(R+d)(\alpha-\theta) + 2\alpha(R+w) - 2\alpha(R+d) \\ &= 2(R+d)(\alpha-\theta) + 2\alpha(w-d), \end{aligned}$$

as required (we added and subtracted $2\alpha(R+d)$).

The approximation. We now work to first order in ε, ignoring terms of order ε^2 relative to those we keep. By (iii), $\alpha-\theta$ is small, so $\alpha-\theta \approx \sin(\alpha-\theta)$, with error of order $(\alpha-\theta)^3$. By (b),

$$2(R+d)(\alpha-\theta) \approx 2(R+d)\cdot\frac{d\sin\alpha}{R+d} = 2d\sin\alpha.$$

Hence

$$S \approx 2d\sin\alpha + 2\alpha w - 2\alpha d = 2\alpha w - 2d(\alpha-\sin\alpha).$$

Now use $d \approx \dfrac{w}{1-\cos\alpha}$ from (iii):

$$S \approx 2w\left(\alpha - \frac{\alpha-\sin\alpha}{1-\cos\alpha}\right) = 2w\cdot\frac{\alpha-\alpha\cos\alpha-\alpha+\sin\alpha}{1-\cos\alpha} = \frac{2w(\sin\alpha-\alpha\cos\alpha)}{1-\cos\alpha}.$$

Finally $L = 2\alpha R(1+\varepsilon) \approx 2\alpha R$, and since S is already of first order in ε, replacing L by $2\alpha R$ only changes S/L at second order. So

$$\frac{S}{L} \approx \left(\frac{\sin\alpha-\alpha\cos\alpha}{\alpha(1-\cos\alpha)}\right)\frac{w}{R},$$

which is the required fraction.

Checks

- At $\alpha = \frac{1}{2}\pi$ the fraction is $\frac{2}{\pi} \cdot \frac{w}{R} \approx 0.637\,\frac{w}{R}$; at $\alpha = \frac{1}{3}\pi$ it is about $0.654\,\frac{w}{R}$. Both are positive, as they must be, since $\tan\alpha > \alpha$ gives $\sin\alpha > \alpha\cos\alpha$.
- With $R = 1$, $w = 0.01$, $\alpha = \frac{1}{2}\pi$: the exact formulas give $d \approx 0.01005$, $\alpha - \theta \approx 0.00995$ and $S/L \approx 0.006285$, against the approximation 0.006366. The relative error, about 1%, is of order w/R, as it should be, and it shrinks tenfold when w is divided by ten.
- A computer check placed A and B in coordinates, found the tangent circle directly, and confirmed the formulas for d, $\sin(\alpha - \theta)$ and S for thousands of random choices of R, w and α.

Common slips

- Taking $OO' = R + d + R$ or $OO' = w$: the circles touch *internally*, so the distance between centres is the difference of the radii, d.
- Using $\cos\alpha$ instead of $\cos(\pi - \alpha)$ for the angle at O in triangle AOO', which flips the sign of the last term.
- In (iii), saying only that $\sin(\alpha - \theta)$ is small: you also need $\alpha - \theta$ to lie in $(0, \frac{1}{2}\pi)$ before you can conclude that the angle itself is small.

Beyond STEP

"Much less than 1" is the language of perturbation theory: expand in a small parameter ε, keep the leading term, and make sure that the error really is of higher order. At university this is made precise with Taylor's theorem and big-O notation, and here it would show that S/L equals the given fraction times ε, plus a term of order ε^2. For a careful treatment of Taylor's theorem with remainder, see Abbott, *Understanding Analysis*; for more problems in the same spirit, see Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP II 2021, Question 7

Pure • Rotation Matrices

STEP II 2021, Question 7

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

The key idea

A 2×2 rotation matrix has a very rigid shape: it is $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$, so its columns are unit vectors. Asking that $\mathbf{R} + \mathbf{I}$ keeps this shape pins down ϕ completely. For a general matrix with $\mathbf{S}^3 = \mathbf{I}$, the useful tools are the determinant (which is multiplicative) and the relation $\mathbf{S}^2 = (a+d)\mathbf{S} - \mathbf{I}$, which lets us write any power of $\mathbf{S}$ as a combination of $\mathbf{S}$ and $\mathbf{I}$.

Solution

Part (i)

Write $c = \cos\phi$ and $s = \sin\phi$, so

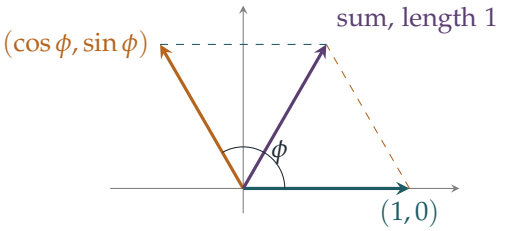
$$\mathbf{R} = \begin{pmatrix} c & -s \\ s & c \end{pmatrix}, \qquad \mathbf{R} + \mathbf{I} = \begin{pmatrix} 1+c & -s \\ s & 1+c \end{pmatrix}.$$

This already has the pattern $\begin{pmatrix} p & -q \\ q & p \end{pmatrix}$, and such a matrix is a rotation exactly when $p^2 + q^2 = 1$ (then $(p, q) = (\cos\theta, \sin\theta)$ for some θ). So we need

$$(1+c)^2 + s^2 = 1 \iff 2 + 2c = 1 \iff \cos\phi = -\tfrac{1}{2}.$$

With $0^\circ \leq \phi < 360^\circ$ this gives $\phi = 120^\circ$ or $\phi = 240^\circ$.

Geometrically, the first column of $\mathbf{R} + \mathbf{I}$ is the sum of the unit vectors $(1, 0)$ and $(\cos\phi, \sin\phi)$, and we need this sum to be a unit vector too: the three vectors form an equilateral triangle, as in the picture for $\phi = 120^\circ$.



In either case $\mathbf{R}^3$ is a rotation through 3ϕ, which is $360°$ or $720°$, a whole number of turns. Hence $\mathbf{R}^3 = \mathbf{I}$.

Part (ii)

The determinant. Since det is multiplicative, $(\det \mathbf{S})^3 = \det(\mathbf{S}^3) = \det \mathbf{I} = 1$. The matrix is real, so $\det \mathbf{S}$ is a real number whose cube is 1, and the only such number is 1. So $\det \mathbf{S} = ad - bc = 1$.

The quadratic relation. Multiplying out,

$$\mathbf{S}^2 = \begin{pmatrix} a^2 + bc & ab + bd \\ ca + dc & bc + d^2 \end{pmatrix}, \qquad (a+d)\mathbf{S} - \mathbf{I} = \begin{pmatrix} a^2 + ad - 1 & ab + bd \\ ac + dc & ad + d^2 - 1 \end{pmatrix}.$$

The off-diagonal entries agree, and the diagonal entries agree because $bc = ad - 1$ (that is, $\det \mathbf{S} = 1$). Hence $\mathbf{S}^2 = (a+d)\mathbf{S} - \mathbf{I}$, as required.

The trace. Let $t = a + d$. Multiply the relation by $\mathbf{S}$ and use it again:

$$\mathbf{S}^3 = t\mathbf{S}^2 - \mathbf{S} = t\big(t\mathbf{S} - \mathbf{I}\big) - \mathbf{S} = (t^2 - 1)\mathbf{S} - t\mathbf{I}.$$

Setting $\mathbf{S}^3 = \mathbf{I}$ gives

$$(t^2 - 1)\mathbf{S} = (t+1)\mathbf{I}.$$

> **Ask yourself** Can we just divide by $t^2 - 1$ and be done?

Not without care: we must look at the case $t^2 - 1 = 0$ separately, and in fact it is the case that happens.

- If $t^2 \neq 1$, then $\mathbf{S} = \dfrac{t+1}{t^2-1}\mathbf{I} = k\mathbf{I}$ with k real. Then $\mathbf{S}^3 = k^3\mathbf{I} = \mathbf{I}$ forces $k^3 = 1$, so $k = 1$ and $\mathbf{S} = \mathbf{I}$, which is excluded.
- So $t^2 = 1$. The equation becomes $\mathbf{0} = (t+1)\mathbf{I}$, so $t = -1$ (the value $t = 1$ would give $2\mathbf{I} = \mathbf{0}$, which is false).

Therefore $a + d = -1$.

Part (iii)

First, $\mathbf{S} \neq \mathbf{I}$: otherwise $\mathbf{S} + \mathbf{I} = 2\mathbf{I}$, which has determinant 4 and doubles lengths, so is not a rotation. Part (ii) therefore applies, and $a + d = -1$.

Since $\mathbf{S}+\mathbf{I}$ is a rotation, say through θ,

$$\mathbf{S}=\begin{pmatrix}\cos\theta-1 & -\sin\theta\\ \sin\theta & \cos\theta-1\end{pmatrix}.$$

The trace condition gives $2\cos\theta-2=-1$, so $\cos\theta=\frac{1}{2}$ and $\sin\theta=\pm\frac{\sqrt{3}}{2}$. Then

$$\mathbf{S}=\begin{pmatrix}-\frac{1}{2} & -\sin\theta\\ \sin\theta & -\frac{1}{2}\end{pmatrix},$$

which has the rotation pattern with $\left(-\frac{1}{2}\right)^2+\sin^2\theta=\frac{1}{4}+\frac{3}{4}=1$. So $\mathbf{S}$ is a rotation, with cosine $-\frac{1}{2}$ and sine $\pm\frac{\sqrt{3}}{2}$.

The angle is 120° (when $\theta=60^\circ$) or 240° (when $\theta=300^\circ$). Both really occur: by part (i), each of these rotations has $\mathbf{S}+\mathbf{I}$ a rotation and $\mathbf{S}^3=\mathbf{I}$. (Equivalently, once we know $\mathbf{S}$ is a rotation, part (i) itself lists the possible angles.)

Checks

- With $\phi=120^\circ$, $\mathbf{R}+\mathbf{I}=\begin{pmatrix}\frac{1}{2} & -\frac{\sqrt{3}}{2}\\ \frac{\sqrt{3}}{2} & \frac{1}{2}\end{pmatrix}$, which is the rotation through 60°; with $\phi=240^\circ$ it is the rotation through 300°.
- The matrix $\mathbf{S}=\begin{pmatrix}0 & -1\\ 1 & -1\end{pmatrix}$ has $\mathbf{S}^3=\mathbf{I}$ (multiply it out) and is not a rotation; it still has determinant 1 and trace -1, as part (ii) predicts. Its $\mathbf{S}+\mathbf{I}=\begin{pmatrix}1 & -1\\ 1 & 0\end{pmatrix}$ is not a rotation, consistent with part (iii).
- A computer check of many random real matrices with $\mathbf{S}^3=\mathbf{I}$, $\mathbf{S}\neq\mathbf{I}$ confirmed determinant 1, trace -1 and the relation for $\mathbf{S}^2$ every time; scanning θ in part (iii) found only $\theta=60^\circ,300^\circ$.

Common slips

- Dividing by t^2-1 without checking whether it can be zero; the case $t^2=1$ is exactly the answer.
- Forgetting that $\mathbf{S}\neq\mathbf{I}$ must be established in part (iii) before part (ii) can be used.
- Thinking $\det(\mathbf{S}+\mathbf{I})=1$ is enough to make $\mathbf{S}+\mathbf{I}$ a rotation; here it holds automatically, so it gives no information.

Beyond STEP

The relation $\mathbf{S}^2=(a+d)\mathbf{S}-(\det\mathbf{S})\mathbf{I}$ is the 2×2 case of the Cayley–Hamilton theorem: every square matrix satisfies its own characteristic polynomial. Part (ii) says that a real

2×2 matrix of order 3 has eigenvalues the two non-real cube roots of unity, and such a matrix is always *similar* to a rotation through $120°$, though not necessarily equal to one. Strang's *Introduction to Linear Algebra* develops eigenvalues and similarity.

WORKED SOLUTION

STEP II 2021, Question 8

Pure • Integral Recurrence Relations

STEP II 2021, Question 8

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 2.9, *Fractions That Close In on e*: https://beyondstep.keithmcnulty.org/ch02/e-approximations.html

The key idea

The function $t^n(1-t)^n$ is a power of the single quadratic $u = t(1-t)$, so differentiating it is just the chain rule twice. That function and its first derivative both vanish at $t = 0$ and $t = 1$, so integrating by parts twice against e^t throws away every boundary term and turns the second derivative from part (i) straight into a recurrence. The integrals are squeezed between 0 and something tiny, and this forces e to be very well approximated by the fractions $-a_n/b_n$.

Solution

Part (i)

Let $u = t(1-t) = t - t^2$, so that $\dfrac{\mathrm{d}u}{\mathrm{d}t} = 1 - 2t$, $\dfrac{\mathrm{d}^2u}{\mathrm{d}t^2} = -2$, and

$$(1-2t)^2 = 1 - 4t + 4t^2 = 1 - 4u.$$

With $f(t) = t^n(1-t)^n = u^n$, the chain rule gives $f'(t) = nu^{n-1}(1-2t)$. Differentiating again by the product rule (valid for $n \geq 2$, so that u^{n-2} is a polynomial),

$$\begin{aligned} f''(t) &= n(n-1)u^{n-2}(1-2t)^2 - 2nu^{n-1} \\ &= nu^{n-2}\big[(n-1)(1-4u) - 2u\big] \\ &= nu^{n-2}\big[(n-1) - (4n-2)u\big] \\ &= nt^{n-2}(1-t)^{n-2}\big[(n-1) - 2(2n-1)t(1-t)\big], \end{aligned}$$

which is the required result.

Part (ii)

Keep $f(t) = t^n(1-t)^n$ with $n \geq 2$, so $n!\,T_n = \int_0^1 f(t)\,\mathrm{e}^t\,\mathrm{d}t$. Integrating by parts twice, each time differentiating the polynomial and integrating e^t,

$$\int_0^1 f\,\mathrm{e}^t\,\mathrm{d}t = \Big[f\,\mathrm{e}^t\Big]_0^1 - \Big[f'\,\mathrm{e}^t\Big]_0^1 + \int_0^1 f''\,\mathrm{e}^t\,\mathrm{d}t.$$

? **Ask yourself** Why do both boundary terms disappear?

Because $f(t) = u^n$ and $f'(t) = nu^{n-1}(1-2t)$ both contain a factor $u = t(1-t)$ (here we use $n \geq 2$, so $n-1 \geq 1$), and $u = 0$ at $t = 0$ and at $t = 1$. So

$$n!\,T_n = \int_0^1 f''(t)\,\mathrm{e}^t\,\mathrm{d}t.$$

Now substitute part (i), writing it as $f'' = n(n-1)u^{n-2} - 2n(2n-1)u^{n-1}$:

$$\begin{aligned} n!\,T_n &= n(n-1)\int_0^1 u^{n-2}\mathrm{e}^t\,\mathrm{d}t - 2n(2n-1)\int_0^1 u^{n-1}\mathrm{e}^t\,\mathrm{d}t \\ &= n(n-1)\,(n-2)!\,T_{n-2} - 2n(2n-1)\,(n-1)!\,T_{n-1} \\ &= n!\,T_{n-2} - 2(2n-1)\,n!\,T_{n-1}, \end{aligned}$$

using $\int_0^1 u^k\mathrm{e}^t\,\mathrm{d}t = k!\,T_k$. Dividing by $n!$ gives

$$T_n = T_{n-2} - 2(2n-1)T_{n-1} \qquad (n \geq 2),$$

as required.

Part (iii)

First, $T_0 = \int_0^1 \mathrm{e}^t\,\mathrm{d}t = \mathrm{e} - 1$.

For $T_1 = \int_0^1 (t - t^2)\mathrm{e}^t\,\mathrm{d}t$ we can check by differentiating that $\int t\,\mathrm{e}^t\,\mathrm{d}t = (t-1)\mathrm{e}^t$ and $\int t^2\mathrm{e}^t\,\mathrm{d}t = (t^2 - 2t + 2)\mathrm{e}^t$, so

$$\int_0^1 (t - t^2)\mathrm{e}^t\,\mathrm{d}t = \Big[(-t^2 + 3t - 3)\mathrm{e}^t\Big]_0^1 = -\mathrm{e} + 3,$$

that is, $T_1 = 3 - \mathrm{e}$.

So $T_0 = a_0 + b_0\mathrm{e}$ with $(a_0, b_0) = (-1, 1)$, and $T_1 = a_1 + b_1\mathrm{e}$ with $(a_1, b_1) = (3, -1)$, all integers. Suppose that T_{n-2} and T_{n-1} have this form with integer coefficients, for some $n \geq 2$. By part (ii),

$$T_n = \big(a_{n-2} - 2(2n-1)a_{n-1}\big) + \big(b_{n-2} - 2(2n-1)b_{n-1}\big)\mathrm{e},$$

and both brackets are integers because $2(2n-1)$ is an integer. By (strong) induction, every T_n with $n \geq 0$ can be written as $a_n + b_n\mathrm{e}$ with $a_n, b_n \in \mathbb{Z}$.

Part (iv)

The bounds. For $0 < t < 1$ we have $0 < t(1-t) \le \frac{1}{4} < 1$, so $0 < t^n(1-t)^n \le 1$, and also $0 < \mathrm{e}^t < \mathrm{e}$. Hence on the open interval $(0,1)$

$$0 < \frac{t^n(1-t)^n}{n!}\,\mathrm{e}^t < \frac{\mathrm{e}}{n!}.$$

The integrand is continuous, so integrating these strict inequalities over $[0,1]$ (an interval of length 1) keeps them strict:

$$0 < T_n < \frac{\mathrm{e}}{n!} \qquad (n \ge 0).$$

(For $n = 0$ the left factor is just 1, and the strictness comes from $\mathrm{e}^t < \mathrm{e}$.)

The limit. Since $b_n \neq 0$, divide $T_n = a_n + b_n\mathrm{e}$ by b_n:

$$-\frac{a_n}{b_n} - \mathrm{e} = -\frac{T_n}{b_n}, \qquad \text{so} \qquad \left|-\frac{a_n}{b_n} - \mathrm{e}\right| = \frac{T_n}{|b_n|} \le T_n < \frac{\mathrm{e}}{n!}.$$

Here we used that b_n is a non-zero *integer*, so $|b_n| \ge 1$. As $n \to \infty$, $\mathrm{e}/n! \to 0$, so $-a_n/b_n \to \mathrm{e}$.

In fact the bound is very generous. From the recurrence, $(a_2, b_2) = (-19, 7)$ and $(a_3, b_3) = (193, -71)$, giving the approximations $\frac{19}{7} \approx 2.714$ and $\frac{193}{71} \approx 2.71831$, against $\mathrm{e} \approx 2.71828$.

Checks

- Put $n = 2$ in part (i): $t^2(1-t)^2 = t^2 - 2t^3 + t^4$ has second derivative $2 - 12t + 12t^2$, and the formula gives $2\big[1 - 6t(1-t)\big] = 2 - 12t + 12t^2$.
- $T_1 = 3 - \mathrm{e} \approx 0.2817$ is positive and less than $\mathrm{e}/1!$, as part (iv) says it must be. Then $T_2 = T_0 - 6T_1 = 7\mathrm{e} - 19 \approx 0.0280$, which a computer evaluation of the integral confirms, along with the recurrence and the bounds for n up to 11.
- The values $-a_n/b_n$ for $n = 4, 5, 6$ are $\frac{2721}{1001}$, $\frac{49171}{18089}$, $\frac{1084483}{398959}$, which differ from e by about 10^{-7}, 3×10^{-10} and 5×10^{-13}: much faster than $\mathrm{e}/n!$ alone would suggest.

Common slips

- Forgetting to say why the boundary terms vanish, or not noticing that this needs $n \ge 2$: for $n = 1$ the derivative $1 - 2t$ is non-zero at both ends.
- Using only $b_n \neq 0$ in part (iv) without $|b_n| \ge 1$. A non-zero real number could be tiny; it is the fact that b_n is an integer that stops T_n/b_n from blowing up.

Beyond STEP

This is the heart of Hermite's 1873 approach to proving that e is irrational and even transcendental: build integer combinations $a_n + b_n\mathrm{e}$ that are non-zero but tend to zero. If $\mathrm{e} = p/q$, then $q(a_n + b_n\mathrm{e}) = qa_n + b_n p$ would be a positive integer less than $q\mathrm{e}/n!$, which is impossible for large n; the same polynomials $t^n(1-t)^n$ appear in Beukers' famous proof that $\zeta(3)$ is irrational. For more on how well numbers can be approximated by fractions, read Davenport, *The Higher Arithmetic*.

WORKED SOLUTION

STEP II 2021, Question 9

Mechanics • Statics

STEP II 2021, Question 9

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2021 STEP papers (a zip file):

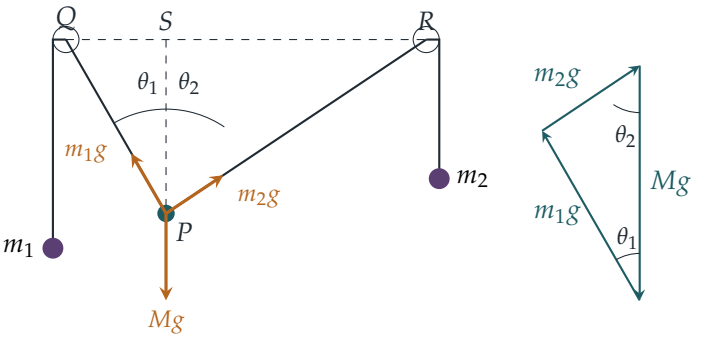
www.ocr.org.uk/Images/696588-2021-past-papers.zip

The key idea

Because the pulleys are smooth, each string section carries the weight of the particle hanging at its far end, so the three forces on P have magnitudes in the ratio $m_1 : m_2 : M$. Those three forces close up into a triangle whose sides are $m_1 g$, $m_2 g$ and Mg, and whose angles are built from θ_1 and θ_2. The cosine rule on that triangle then gives everything: the inequalities come from asking that the triangle exists with both angles acute, and the ratio comes from $\tan\theta_1$ and $\tan\theta_2$.

Solution

Model. The particles are point masses, the string is light and inextensible, and the pulleys are small and smooth. So the tension in PQ equals the tension in the vertical part supporting m_1, which is in equilibrium: $T_1 = m_1 g$. Likewise the tension in PR is $T_2 = m_2 g$. Write $h = SP$.



Part (i)

Resolving horizontally and vertically at P (and cancelling g):

$$m_1 \sin\theta_1 = m_2 \sin\theta_2, \qquad m_1 \cos\theta_1 + m_2 \cos\theta_2 = M. \tag{$*$}$$

Equivalently, the three forces placed nose to tail form a closed triangle with sides $m_1 g$, $m_2 g$, Mg (right of the figure). The side Mg is vertical, so the angle between the sides Mg and $m_1 g$ is θ_1 (opposite the side $m_2 g$), and the angle between Mg and $m_2 g$ is θ_2 (opposite $m_1 g$). The third angle, opposite Mg, is $180^\circ - \theta_1 - \theta_2$.

If you prefer algebra to the picture, the same facts drop out of $(*)$: write $m_1 \cos\theta_1 = M - m_2 \cos\theta_2$ and $m_1 \sin\theta_1 = m_2 \sin\theta_2$, square and add:

$$m_1^2 = M^2 + m_2^2 - 2Mm_2 \cos\theta_2, \quad \text{so} \quad \cos\theta_2 = \frac{M^2 + m_2^2 - m_1^2}{2Mm_2},$$

and, swapping the roles of the two sides,

$$\cos\theta_1 = \frac{M^2 + m_1^2 - m_2^2}{2Mm_1}.$$

(a) The triangle of forces is a genuine (non-flat) triangle, because $\theta_1 + \theta_2$ lies strictly between 0° and 180°. The triangle inequality is therefore strict, and gives $M < m_1 + m_2$.

For the lower bound we use the fact that θ_2 is acute, so $\cos\theta_2 > 0$. Since $2Mm_2 > 0$, the formula above gives

$$M^2 + m_2^2 - m_1^2 > 0 \iff M^2 > m_1^2 - m_2^2 \iff M > \sqrt{m_1^2 - m_2^2},$$

where the last step is valid because $M > 0$ and $m_1^2 - m_2^2 > 0$. Together,

$$\sqrt{m_1^2 - m_2^2} < M < m_1 + m_2.$$

? **Ask yourself** Why does the condition on θ_2 matter, and not the one on θ_1?

In a triangle the larger side faces the larger angle. Since $m_1 > m_2$, the angle θ_2 (opposite $m_1 g$) is larger than θ_1 (opposite $m_2 g$). So once θ_2 is acute, θ_1 is automatically acute too; the only real restriction is on θ_2. (Indeed $\cos\theta_1 > 0$ holds for every $M > 0$, as $m_1 > m_2$.)

(b) In the right-angled triangles QSP and RSP, with $SP = h$,

$$QS = h\tan\theta_1, \qquad SR = h\tan\theta_2.$$

Hence

$$r = \frac{QS}{SR} = \frac{\tan\theta_1}{\tan\theta_2} = \frac{\sin\theta_1}{\sin\theta_2}\cdot\frac{\cos\theta_2}{\cos\theta_1}$$
$$= \frac{m_2}{m_1}\cdot\frac{M^2+m_2^2-m_1^2}{2Mm_2}\cdot\frac{2Mm_1}{M^2+m_1^2-m_2^2} \quad \text{(using } (*) \text{ and the cosines)}$$
$$= \frac{M^2-m_1^2+m_2^2}{M^2-m_2^2+m_1^2},$$

which is the required ratio. Both brackets are positive by part (a), so $r > 0$, and $r < 1$ because $m_1 > m_2$: the point S is nearer Q, as the figure suggests.

Part (ii)

The cosine rule for the side Mg of the triangle of forces, whose opposite angle is $180^\circ - (\theta_1+\theta_2)$, reads

$$M^2 = m_1^2 + m_2^2 + 2m_1m_2\cos(\theta_1+\theta_2).$$

(Algebraically: square and add the two equations in $(*)$.) With $M^2 = m_1^2+m_2^2$ this forces $\cos(\theta_1+\theta_2) = 0$. Since θ_1 and θ_2 are acute, $0^\circ < \theta_1+\theta_2 < 180^\circ$, and the only possibility is

$$\theta_1+\theta_2 = 90^\circ.$$

So the string makes a right angle at P. (This value of M does lie in the range of part (a), so the configuration exists.)

Now the cosines simplify: $\cos\theta_1 = \dfrac{2m_1^2}{2Mm_1} = \dfrac{m_1}{M}$ and $\cos\theta_2 = \dfrac{m_2}{M}$. Since $\theta_2 = 90^\circ - \theta_1$, we get $\sin\theta_1 = \cos\theta_2 = m_2/M$, so

$$\tan\theta_1 = \frac{m_2}{m_1}, \qquad \tan\theta_2 = \frac{1}{\tan\theta_1} = \frac{m_1}{m_2}.$$

Therefore

$$\frac{QR}{SP} = \frac{h\tan\theta_1 + h\tan\theta_2}{h} = \frac{m_2}{m_1}+\frac{m_1}{m_2} = \frac{m_1^2+m_2^2}{m_1m_2},$$

so $QR : SP = (m_1^2+m_2^2) : m_1m_2$.

Checks

- With $m_1 = 5$, $m_2 = 3$, $M = 6$ (inside $4 < M < 8$), solving $(*)$ numerically gives $\theta_1 \approx 29.93^\circ$ and $\theta_2 \approx 56.25^\circ$. Then $\tan\theta_1/\tan\theta_2 \approx 0.3846 = \frac{5}{13}$, which matches $r = \frac{36-25+9}{36-9+25} = \frac{20}{52}$.
- At the ends of the range: as $M \to \sqrt{m_1^2-m_2^2}$ we get $\theta_2 \to 90^\circ$ and $r \to 0$ (the string PR becomes horizontal); as $M \to m_1+m_2$ both angles tend to 0 and P sinks far below QR.
- Part (ii) with $m_1 = 4$, $m_2 = 3$, $M = 5$: $\tan\theta_1 = \frac{3}{4}$, $\tan\theta_2 = \frac{4}{3}$, and $QR/SP = \frac{25}{12}$. Check $(*)$: $4\cdot\frac{3}{5} = 3\cdot\frac{4}{5}$ and $4\cdot\frac{4}{5}+3\cdot\frac{3}{5} = 5$.

Common slips

- Taking the angle opposite Mg in the triangle of forces to be $\theta_1 + \theta_2$ rather than $180^\circ - (\theta_1 + \theta_2)$, which flips the sign in the cosine rule.
- Using a non-strict triangle inequality, or forgetting that the lower bound needs the angles to be acute.
- Inverting the ratio: $r = QS/SR$, and QS is the side next to the heavier particle.

Beyond STEP

The triangle of forces is the simplest case of a general fact: forces in equilibrium sum to the zero vector, so drawn nose to tail they form a closed polygon. Engineers use this idea (the "force polygon") to analyse frameworks, and in first-year mechanics it reappears as the vector form of statics, alongside the condition that moments also balance. For more, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP II 2021, Question 10

Mechanics • Non-Inertial Frames

STEP II 2021, Question 10

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 11.5, *Gravity in an Accelerating Train*: https://beyondstep.keithmcnulty.org/ch11/accelerating-frame.html

The key idea

The axes move with the train, so they are not an inertial frame. The cleanest fix is to work out the bead's *true* acceleration: its acceleration relative to the train plus the train's own acceleration. Newton's second law then holds, and because the wire is smooth the reaction is perpendicular to the wire, so it does no work along the bead's path. Dotting the equation of motion with the bead's (relative) velocity removes the unknown reaction and leaves an energy-like quantity that is conserved.

Solution

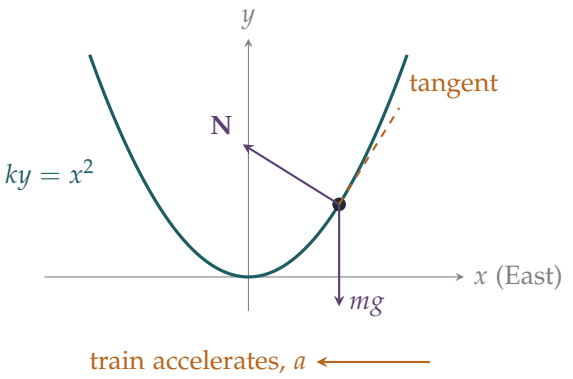
Model the bead as a particle of mass m and the wire as smooth, so the reaction $\mathbf{N}$ of the wire on the bead is always perpendicular to the wire. Write $\mathbf{i}$ and $\mathbf{j}$ for unit vectors east and up.

The train accelerates westwards, so its acceleration is $-a\mathbf{i}$. The bead's position relative to the train is $x\mathbf{i} + y\mathbf{j}$, so its true (ground-frame) acceleration is

$$(\ddot{x} - a)\,\mathbf{i} + \ddot{y}\,\mathbf{j}.$$

The only forces on the bead are its weight $-mg\mathbf{j}$ and $\mathbf{N}$, so Newton's second law gives

$$m(\ddot{x} - a)\,\mathbf{i} + m\ddot{y}\,\mathbf{j} = \mathbf{N} - mg\,\mathbf{j}. \qquad (*)$$



Part (i)

Suppose the bead stayed at the origin relative to the train. Then it would share the train's acceleration, $-a\mathbf{i}$, so it would need a horizontal force $-ma\mathbf{i}$ with $a > 0$. But at the origin the wire $y = x^2/k$ has gradient $2x/k = 0$, so it is horizontal there, and the reaction from a smooth wire is perpendicular to it, which means vertical. Gravity is vertical too. No horizontal force acts, so the bead cannot accelerate with the train; it must move relative to the train. (Its true horizontal acceleration is 0 while the train's is a westwards, so the bead starts to slide east along the wire, towards positive x.)

Part (ii)

While the bead moves along the wire, its velocity relative to the train, $\dot{x}\,\mathbf{i} + \dot{y}\,\mathbf{j}$, is tangent to the wire, because the wire is fixed in the train. Since $\mathbf{N}$ is perpendicular to the wire,

$$\mathbf{N} \cdot (\dot{x}\,\mathbf{i} + \dot{y}\,\mathbf{j}) = 0.$$

Rearranging $(*)$ as $m(\ddot{x} - a)\,\mathbf{i} + m(\ddot{y} + g)\,\mathbf{j} = \mathbf{N}$ and taking the scalar product of both sides with $\dot{x}\,\mathbf{i} + \dot{y}\,\mathbf{j}$ gives, after dividing by m,

$$\dot{x}(\ddot{x} - a) + \dot{y}(\ddot{y} + g) = 0,$$

as required. The left-hand side is an exact derivative:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\tfrac{1}{2}(\dot{x}^2 + \dot{y}^2) - ax + gy\right) = \dot{x}\ddot{x} + \dot{y}\ddot{y} - a\dot{x} + g\dot{y} = 0,$$

so $\frac{1}{2}(\dot{x}^2 + \dot{y}^2) - ax + gy$ is constant during the motion.

Ask yourself Why is it legitimate to dot with the *relative* velocity, when the reaction does work in the ground frame?

In the ground frame the bead's velocity has an extra $-at\,\mathbf{i}$, which is not along the wire, so $\mathbf{N}$ does work there. But the identity $\mathbf{N} \cdot \mathbf{u} = 0$ is pure geometry: it holds for any vector $\mathbf{u}$ tangent to the wire, and the relative velocity is such a vector. That is all we used.

Part (iii)

Initially $x = y = 0$ and $\dot{x} = \dot{y} = 0$, so the constant is 0. Writing v for the speed relative to the train and using $y = x^2/k$,

$$\tfrac{1}{2}v^2 = ax - gy = ax - \frac{gx^2}{k} = \frac{g}{k}\,x\Big(\frac{ak}{g} - x\Big). \qquad (\dagger)$$

Since $v^2 \geq 0$, the bead is confined to $0 \leq x \leq ak/g$, and $v = 0$ only at the two ends of this interval. By part (i) the bead starts moving towards positive x. While $0 < x < ak/g$ its speed is positive, so it keeps moving along the wire in the same direction until it reaches $x = ak/g$, where it momentarily stops. (It does turn back there: the tangential component of $(*)$ at that point is not zero, since $\frac{\mathrm{d}}{\mathrm{d}x}(ax - gx^2/k) = a - 2a = -a \neq 0$.) The bead then oscillates between $x = 0$ and $x = ak/g$.

On this interval $y = x^2/k$ is increasing in x, so the greatest height is reached at $x = ak/g$:

$$b = \frac{1}{k}\Big(\frac{ak}{g}\Big)^2 = \frac{a^2k}{g^2}.$$

Part (iv)

By $(\dagger)$, $v^2 = 2\big(ax - gx^2/k\big)$, a quadratic in x with a maximum at the midpoint of its roots 0 and ak/g, so the speed relative to the train is greatest when

$$x = \frac{ak}{2g}.$$

There

$$v^2 = 2\Big(\frac{a^2k}{2g} - \frac{a^2k}{4g}\Big) = \frac{a^2k}{2g}, \qquad \text{so} \qquad v_{\max} = a\sqrt{\frac{k}{2g}}.$$

Checks

- Limiting case: if $a = 0$ the bead simply stays at the bottom of the wire, and indeed $b = 0$ and $v_{\max} = 0$.
- Effective gravity: in the train the bead feels an effective gravity $(a, -g)$, tilted towards the east. The point of fastest motion, $x = ak/(2g)$, is exactly where the wire's gradient $2x/k$ equals a/g, so the tangent is perpendicular to the effective gravity: the "lowest point" of the wire in the tilted field. The turning point $x = ak/g$ is twice as far out, as for a symmetric oscillation in a potential that is quadratic in x.
- Dimensions: k is a length, so a^2k/g^2 is a length and $a\sqrt{k/g}$ is a speed.
- Numerically, with $a = 2$, $k = 1.5$ and $g = 9.8$ (SI units), stepping the motion forward by computer gives a greatest height of $0.06247\,\mathrm{m}$ and a greatest speed of $0.5533\,\mathrm{m\,s^{-1}}$ at $x = 0.1531\,\mathrm{m}$, matching a^2k/g^2, $a\sqrt{k/(2g)}$ and $ak/(2g)$.

Common slips

- Writing $\ddot{x}$ for the bead's true horizontal acceleration. That is its acceleration relative to the train; the train's own acceleration $-a$ must be added.
- Getting the sign of a wrong: the train accelerates *west*, so in the train the bead is pushed *east*, towards positive x.
- Taking the speed to be greatest at the bottom of the wire, $x = 0$. In the train the effective "down" is tilted, so the fastest point is at $x = ak/(2g)$ instead.

Beyond STEP

Part (ii) is a first taste of mechanics in non-inertial frames: in the train's frame the term $-ax$ acts like a potential energy for a uniform "fictitious" force ma eastwards, and adding it to gravity gives a tilted effective gravity. At university this extends to rotating frames, where centrifugal and Coriolis forces appear, and to Lagrangian mechanics, where constraint forces like $\mathbf{N}$ are eliminated from the start. Kleppner and Kolenkow, *An Introduction to Mechanics*, has a good chapter on non-inertial systems and fictitious forces.

WORKED SOLUTION

STEP II 2021, Question 11

Probability and Statistics • Conditional Probability Recurrence

STEP II 2021, Question 11

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 12.5, *The Last Passenger's Seat*: https://beyondstep.keithmcnulty.org/ch12/airline-seat.html

The key idea

Condition on where T_1 sits. If T_1 takes their own seat S_1, everyone else is fine; if T_1 takes S_k, then $T_2, \ldots, T_{k-1}$ sit happily and T_k becomes a new "random chooser". The crucial observation is that T_k is then in exactly the same position as T_1 was, but in a smaller train, with the vacant seat S_1 playing the part of T_k's own seat. That turns the problem into a recurrence.

Solution

Throughout, "T_r is displaced" means T_r arrives to find S_r already taken. Only a displaced passenger (or T_1) ever chooses at random.

Part (i)

With $n = 2$, T_1 takes S_1 or S_2 with probability $\frac{1}{2}$ each, and T_2 gets S_2 exactly when T_1 took S_1. So

$$P_2 = \tfrac{1}{2}.$$

With $n = 3$, condition on T_1's seat, each with probability $\frac{1}{3}$:

- S_1: T_2 and T_3 both find their seats free, so T_3 gets S_3.
- S_3: T_3 certainly does not get S_3.
- S_2: T_2 is displaced and chooses between S_1 and S_3 at random; T_3 gets S_3 exactly when T_2 picks S_1, with probability $\frac{1}{2}$.

Hence

$$P_3 = \tfrac{1}{3}\cdot 1 + \tfrac{1}{3}\cdot 0 + \tfrac{1}{3}\cdot \tfrac{1}{2} = \tfrac{1}{2}.$$

Part (ii)

Suppose T_1 sits in S_k with $2 \le k \le n-1$. Then $T_2, \ldots, T_{k-1}$ all find their seats free and take them. When T_k arrives, the free seats are

$$S_1,\ S_{k+1},\ S_{k+2},\ \ldots,\ S_n,$$

which is $n-k+1$ seats, and T_k chooses among them uniformly at random. The passengers still to come are $T_{k+1}, \ldots, T_n$, each wanting their own seat, and nobody still to come is allocated S_1.

? **Ask yourself** Why is this exactly the original problem with $n-k+1$ seats?

Relabel: call T_k the "first passenger" and pretend S_1 is T_k's allocated seat. In the original problem with m seats, the first passenger chooses uniformly from all m seats, each later passenger takes their own seat if free and otherwise chooses uniformly from the free ones, and the first passenger's allocated seat is wanted by no one else. Here, with $m = n-k+1$, the passengers $T_k, T_{k+1}, \ldots, T_n$ follow precisely those rules with the seats $S_1, S_{k+1}, \ldots, S_n$, and S_1 is wanted by no one else. The rules never use the seat labels except to decide whether a passenger's own seat is free, so the two processes have identical probabilities. In the smaller problem T_n is still the last passenger, so

$$\mathrm{P}(T_n \text{ sits in } S_n \mid T_1 \text{ sits in } S_k) = P_{n-k+1}.$$

Now, for $n \ge 3$, condition on T_1's seat, each of $S_1, \ldots, S_n$ having probability $\frac{1}{n}$. If it is S_1, T_n certainly gets S_n; if it is S_n, T_n certainly does not. So, by the law of total probability,

$$P_n = \frac{1}{n}\cdot 1 + \frac{1}{n}\sum_{k=2}^{n-1} P_{n-k+1} + \frac{1}{n}\cdot 0 = \frac{1}{n}\left(1 + \sum_{r=2}^{n-1} P_r\right),$$

where we substituted $r = n-k+1$, which runs from $n-1$ down to 2 as k runs from 2 to $n-1$. This is the required result.

Part (iii)

We claim that $P_n = \frac{1}{2}$ for every $n \ge 2$. We use strong induction.

Base case. $P_2 = \frac{1}{2}$ from part (i).

Inductive step. Let $n \ge 3$ and suppose $P_r = \frac{1}{2}$ for $r = 2, 3, \ldots, n-1$. The sum in part (ii) has $n-2$ terms, each equal to $\frac{1}{2}$, so

$$P_n = \frac{1}{n}\left(1 + \frac{n-2}{2}\right) = \frac{1}{n}\cdot\frac{n}{2} = \frac{1}{2}.$$

Hence $P_r = \frac{1}{2}$ for $r = 2, \ldots, n$, and by induction $P_n = \frac{1}{2}$ for all $n \ge 2$. (Part (i)'s $P_3 = \frac{1}{2}$ agrees.)

Part (iv)

For $n = 2$, $T_{n-1} = T_1$, who sits in S_1 with probability $\frac{1}{2}$. So $Q_2 = \frac{1}{2}$.

Now let $n \geq 3$, so T_{n-1} is not T_1, and again condition on T_1's seat S_k:

- $k = 1$: everyone gets their own seat, so T_{n-1} gets S_{n-1}.
- $k = n$: $T_2, \ldots, T_{n-1}$ all find their seats free, so again T_{n-1} gets S_{n-1}.
- $k = n - 1$: T_{n-1} certainly does not get S_{n-1}.
- $2 \leq k \leq n - 2$: exactly as in part (ii), the rest of the journey is the original problem with $m = n - k + 1$ seats, led by T_k. In that smaller train T_{n-1} is the second-to-last passenger, so the probability is Q_{n-k+1}. Here m runs over $3, 4, \ldots, n - 1$.

Therefore, for $n \geq 3$,

$$Q_n = \frac{1}{n}\left(2 + \sum_{m=3}^{n-1} Q_m\right).$$

For $n = 3$ the sum is empty and $Q_3 = \frac{2}{3}$. If $Q_m = \frac{2}{3}$ for $3 \leq m \leq n - 1$, then

$$Q_n = \frac{1}{n}\left(2 + \frac{2(n-3)}{3}\right) = \frac{1}{n} \cdot \frac{2n}{3} = \frac{2}{3},$$

so by strong induction $Q_n = \frac{2}{3}$ for every $n \geq 3$. In summary,

$$Q_2 = \tfrac{1}{2}, \qquad Q_n = \tfrac{2}{3} \quad (n \geq 3).$$

Ask yourself Is there a way to see $\frac{1}{2}$ and $\frac{2}{3}$ without any recurrence?

Yes. Whenever someone chooses at random, S_1 and each still-free seat of a later passenger are treated alike, so until the process ends the seats S_1 and S_n are equally likely to be the one taken first; T_n succeeds exactly when S_1 goes first, giving $\frac{1}{2}$. For T_{n-1} (with $n \geq 3$), look at which of S_1, S_{n-1}, S_n is taken first by a random chooser; each is equally likely, and T_{n-1} is displaced only if it is S_{n-1}, giving $1 - \frac{1}{3} = \frac{2}{3}$.

Checks

- The case $n = 3$ can be listed by hand: the outcomes for (T_1, T_2, T_3) are (S_1, S_2, S_3) with probability $\frac{1}{3}$, (S_3, S_2, S_1) with probability $\frac{1}{3}$, and (S_2, S_1, S_3), (S_2, S_3, S_1) with probability $\frac{1}{6}$ each. So T_3 gets S_3 with probability $\frac{1}{2}$ and T_2 gets S_2 with probability $\frac{2}{3}$, as found.
- Exact enumeration of every possible journey by computer for $n = 2, \ldots, 8$ gives $P_n = \frac{1}{2}$ throughout, $Q_n = \frac{2}{3}$ for $n \geq 3$, and confirms the conditional probability in part (ii) for every k. A simulation of 10^5 journeys with $n = 20$ gave 0.500 and 0.668.

Common slips

- Forgetting that T_1 can choose S_1 itself, which gives the "$1+$" in the recurrence.
- In part (iv), missing the case where T_1 takes S_n: this is good news for T_{n-1}, so the constant becomes 2, not 1.
- Applying the Q recurrence at $n = 2$, where T_{n-1} is T_1 and the answer is $\frac{1}{2}$, not $\frac{2}{3}$.

Beyond STEP

The recurrence here is a first-step analysis: condition on the first random choice and recognise a smaller copy of the same problem. The same technique gives hitting probabilities and expected hitting times for Markov chains, a central topic in first-year probability. The symmetry argument above is a taste of how a well-chosen "which happens first?" question can replace a long calculation. Grimmett and Welsh, *Probability: An Introduction*, develops both ideas.

WORKED SOLUTION

STEP II 2021, Question 12

Probability and Statistics • Gambler's Ruin

STEP II 2021, Question 12

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

The key idea

Each match keeps returning to a position it has been in before. So we don't add up an infinite list of ways to win. We let P be the probability of winning from that position, condition on what happens next, and get an equation in which P appears on both sides. The same trick, applied to the cautious game, turns it into the match of part (ii).

Solution

Part (i)

Let P be the probability that A wins the match. Condition on the first game. A wins it with probability p_{A} and the match is over. B wins it with probability p_{B} and A has lost. Otherwise the game is drawn, with probability $1 - p_{\mathrm{A}} - p_{\mathrm{B}}$, and since games are independent the match starts afresh, so A's chance from there is again P. Hence

$$P = p_{\mathrm{A}} + (1 - p_{\mathrm{A}} - p_{\mathrm{B}})P \quad \Longrightarrow \quad (p_{\mathrm{A}} + p_{\mathrm{B}})P = p_{\mathrm{A}} \quad \Longrightarrow \quad P = \frac{p_{\mathrm{A}}}{p_{\mathrm{A}} + p_{\mathrm{B}}},$$

dividing by $p_{\mathrm{A}} + p_{\mathrm{B}} > 0$. (Equivalently, A wins at game k with probability $(1 - p_{\mathrm{A}} - p_{\mathrm{B}})^{k-1} p_{\mathrm{A}}$, and the geometric series converges because $0 < p_{\mathrm{A}} + p_{\mathrm{B}} < 1$.)

Part (ii)

Let D be A's number of wins minus B's after some games. Each game changes D by $+1$ or -1, and the match ends the first time $|D| = 2$.

Even length. After n games, with a wins for A and b for B, we have $a + b = n$ and $D = a - b = n - 2b$, so D has the same parity as n. The match ends when $D = \pm 2$, which is even, so n is even.

Probability. Look at the games in pairs. The match is level ($D = 0$) at the start, and since it can only end after an even number of games, it is level again after every pair that doesn't finish it. From a level position, the next two games are AA with probability

p^2 (A wins), BB with probability q^2 (B wins), or AB or BA with total probability $2pq$, which leaves the match level again. Let P be the probability that A wins from a level position. Then

$$P = p^2 + 2pq\,P \quad \Longrightarrow \quad P = \frac{p^2}{1-2pq}.$$

Since $p+q=1$, $1=(p+q)^2 = p^2+2pq+q^2$, so $1-2pq = p^2+q^2$ and

$$P = \frac{p^2}{p^2+q^2},$$

as required. Here $p^2+q^2>0$, so the division is fine.

Part (iii)

Cautious version. The number of tokens goes up or down by one each round, starting at 1 and stopping at 0 (loss) or 4 (win). It can't jump past 4, so the player wins exactly when the count reaches 4 before 0.

> **Ask yourself** From 2 tokens, what does the player need? Have we met this before?

From 2 tokens the player wins by getting two more round wins than losses (reaching 4) before getting two more losses than wins (reaching 0). That is the match of part (ii), with the player as A and the game of chance as B. So from 2 tokens the probability of winning is $\dfrac{p^2}{p^2+q^2}$.

From 1 token, the first round must be won, or the player drops to 0 and loses. So the first round takes the player to 2 tokens with probability p, and the rounds are independent, so

$$P_{\text{cautious}} = p \cdot \frac{p^2}{p^2+q^2} = \frac{p^3}{p^2+q^2}.$$

(If you prefer equations: with P_k the probability of winning from k tokens, $P_1 = pP_2$, $P_2 = pP_3 + qP_1$ and $P_3 = p + qP_2$ give the same answer.)

Bold version. From 1 token the count goes to 2 or 0, and from 2 it goes to 4 or 0. So the player wins only by winning the first two rounds:

$$P_{\text{bold}} = p^2.$$

Comparison. Both probabilities are positive, so we compare their ratio with 1:

$$\frac{P_{\text{cautious}}}{P_{\text{bold}}} = \frac{p}{p^2+q^2}.$$

Since $p^2+q^2>0$, this ratio is greater than 1 if and only if

$$p > p^2+q^2 \iff p-p^2 > q^2 \iff pq > q^2 \iff p > q,$$

dividing by $q > 0$ at the last step (valid because $p < 1$). And $p > q = 1 - p$ exactly when $p > \frac{1}{2}$. In the same way, the ratio is less than 1 exactly when $p < q$, that is, when $p < \frac{1}{2}$. So the cautious version is better when $\frac{1}{2} < p < 1$ and the bold version is better when $0 < p < \frac{1}{2}$. (At $p = \frac{1}{2}$ both give $\frac{1}{4}$.)

Checks

- At $p = \frac{1}{2}$ the cautious game is a fair random walk, and the chance of reaching 4 before 0 from 1 is $\frac{1}{4}$. Our formula gives $\frac{1}{8}/\frac{1}{2} = \frac{1}{4}$, matching $p^2 = \frac{1}{4}$ for the bold game.
- At $p = 0.7$: cautious gives $0.343/0.58 \approx 0.591$ and bold gives 0.49. At $p = 0.3$: cautious gives $0.027/0.58 \approx 0.0466$ and bold gives 0.09. Both agree with a computer check that follows the token distribution round by round.
- In part (ii), $p = \frac{1}{2}$ gives $\frac{1}{2}$, as symmetry demands.

Common slips

- In part (ii), treating "AB" and "BA" as one outcome of probability pq, rather than $2pq$.
- Forgetting to justify that a non-final pair of games always returns the match to level. It does, because the match can't end after an odd number of games.
- In the comparison, multiplying out or dividing without saying the quantities are positive.

Beyond STEP

The cautious game is the *gambler's ruin* problem. It is a random walk with absorbing barriers, and the general answer for starting at k with target N is $\dfrac{1-(q/p)^k}{1-(q/p)^N}$ when $p \neq q$. The result that bold play is best in an unfavourable game, and timid play in a favourable one, goes back to Dubins and Savage's study of optimal gambling. Grimmett and Welsh, *Probability: An Introduction*, treats random walks and gambler's ruin through exactly this kind of first-step conditioning.

WORKED SOLUTION

STEP III 2021, Question 1

Pure • Normals to Parametric Curves

STEP III 2021, Question 1

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

The key idea

Both parts follow the same recipe: differentiate the parametric equations, take the negative reciprocal of $\mathrm{d}y/\mathrm{d}x$ to get the gradient of the normal, and then tidy the equation. The tidying is where all the fun happens. Multiply through to clear the fractions, and the trigonometric terms collapse into a short, symmetric line equation. Once the line is in the form $x\sin\phi + y\cos\phi = \ldots$ or $x\cos\phi + y\sin\phi = \ldots$, its geometry (intercepts, distance from the origin) can be read off at sight.

Solution

Part (i)

Differentiating with respect to t,

$$\frac{\mathrm{d}x}{\mathrm{d}t} = 12\cos^2 t\,\sin t, \quad \frac{\mathrm{d}y}{\mathrm{d}t} = 12\cos t - 12\sin^2 t\cos t = 12\cos t\,(1-\sin^2 t) = 12\cos^3 t.$$

So, for $0 < t < \frac{1}{2}\pi$ (where $\sin t$ and $\cos t$ are nonzero),

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{12\cos^3 t}{12\cos^2 t\sin t} = \frac{\cos t}{\sin t}.$$

At $t = \phi$ the tangent has gradient $\cot\phi$, so the normal has gradient $-\tan\phi = -\dfrac{\sin\phi}{\cos\phi}$.

The normal is therefore

$$y - 12\sin\phi + 4\sin^3\phi = -\frac{\sin\phi}{\cos\phi}\left(x + 4\cos^3\phi\right).$$

Multiply by $\cos\phi$ and collect the x and y terms on the left:

$$\begin{aligned} x\sin\phi + y\cos\phi &= 12\sin\phi\cos\phi - 4\sin^3\phi\cos\phi - 4\cos^3\phi\sin\phi \\ &= 12\sin\phi\cos\phi - 4\sin\phi\cos\phi\,(\sin^2\phi + \cos^2\phi) \\ &= 8\sin\phi\cos\phi. \end{aligned}$$

So the normal is

$$\boxed{x\sin\phi + y\cos\phi = 8\sin\phi\cos\phi} \qquad \text{or equivalently} \qquad \frac{x}{\cos\phi} + \frac{y}{\sin\phi} = 8.$$

The second form shows that the normal meets the axes at $(8\cos\phi, 0)$ and $(0, 8\sin\phi)$.

Tangency. Let $P = (8\cos^3\phi,\ 8\sin^3\phi)$. We need three things.

P is on the curve $x^{2/3} + y^{2/3} = 4$. Since $0 < \phi < \frac{1}{2}\pi$, both coordinates are positive, so the real cube roots are $2\cos\phi$ and $2\sin\phi$. Hence $x^{2/3} + y^{2/3} = 4\cos^2\phi + 4\sin^2\phi = 4$.

P is on the normal. Substituting,

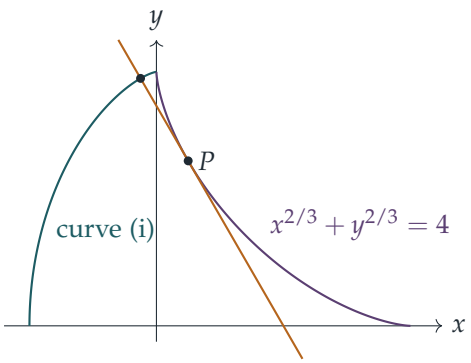
$$8\cos^3\phi\sin\phi + 8\sin^3\phi\cos\phi = 8\sin\phi\cos\phi\,(\cos^2\phi + \sin^2\phi) = 8\sin\phi\cos\phi.$$

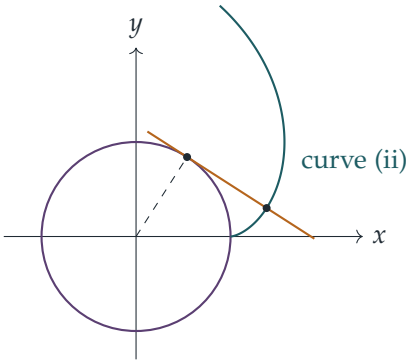
The gradients agree at P. Differentiating $x^{2/3} + y^{2/3} = 4$ implicitly (valid for $x, y > 0$),

$$\tfrac{2}{3}x^{-1/3} + \tfrac{2}{3}y^{-1/3}\frac{\mathrm{d}y}{\mathrm{d}x} = 0 \quad\Longrightarrow\quad \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{y^{1/3}}{x^{1/3}} = -\frac{2\sin\phi}{2\cos\phi} = -\tan\phi \ \text{ at } P,$$

which is exactly the gradient of the normal.

A line through a point of a curve with the same gradient as the curve there is the tangent at that point. So the normal is a tangent to $x^{2/3} + y^{2/3} = 4$ at P, as required.





In each picture the normal (amber) at a point of the curve (teal) touches the second curve (plum); both are drawn for one particular value of ϕ.

Part (ii)

Differentiating with respect to t (product rule on the $t\sin t$ and $t\cos t$ terms),

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -\sin t + \sin t + t\cos t = t\cos t, \qquad \frac{\mathrm{d}y}{\mathrm{d}t} = \cos t - \cos t + t\sin t = t\sin t.$$

For $0 < t < \frac{1}{2}\pi$ we have $t\cos t \neq 0$, so $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{t\sin t}{t\cos t} = \tan t$. At $t = \phi$ the normal has gradient $-\cot\phi = -\dfrac{\cos\phi}{\sin\phi}$, and its equation is

$$y - \sin\phi + \phi\cos\phi = -\frac{\cos\phi}{\sin\phi}\left(x - \cos\phi - \phi\sin\phi\right).$$

Multiplying by $\sin\phi$,

$$y\sin\phi - \sin^2\phi + \phi\sin\phi\cos\phi = -x\cos\phi + \cos^2\phi + \phi\sin\phi\cos\phi.$$

The $\phi\sin\phi\cos\phi$ terms cancel, and $\sin^2\phi + \cos^2\phi = 1$, leaving

$$\boxed{x\cos\phi + y\sin\phi = 1}.$$

Ask yourself How do you find the distance from the origin to a line without finding the foot of the perpendicular?

The distance from (x_0, y_0) to the line $ax + by = c$ is $\dfrac{|ax_0 + by_0 - c|}{\sqrt{a^2+b^2}}$. Here $(x_0, y_0) = (0,0)$, $a = \cos\phi$, $b = \sin\phi$ and $c = 1$, so the distance is

$$\frac{|-1|}{\sqrt{\cos^2\phi + \sin^2\phi}} = 1.$$

You can also see it directly: $(\cos\phi, \sin\phi)$ is a unit vector perpendicular to the line, and the point $(\cos\phi, \sin\phi)$ itself satisfies the equation. So the foot of the perpendicular from O is $(\cos\phi, \sin\phi)$, at distance 1.

The curve. Every normal is at distance 1 from the origin, whatever ϕ is. A line at distance 1 from O meets the circle of radius 1 about O only at the foot of the perpendicular, and there it is perpendicular to the radius. That makes it a tangent to the circle. So every normal is a tangent to

$$\boxed{x^2 + y^2 = 1},$$

touching it at $(\cos\phi, \sin\phi)$.

Checks

- In (i) take $\phi = \frac{1}{3}\pi$. The curve point is $(-\frac{1}{2},\ 6\sqrt{3} - \frac{3}{2}\sqrt{3}) = (-0.5,\ 7.794)$ and the normal $2x + \frac{2}{\sqrt{3}}y = 8$ meets the axes at $(4,0)$ and $(0, 4\sqrt{3}) \approx (0, 6.928)$. The point $P = (1,\ 3\sqrt{3}) \approx (1,\ 5.196)$ satisfies $2 + 6 = 8$, so it lies on the normal, and $1^{2/3} + (3\sqrt{3})^{2/3} = 1 + 3 = 4$.
- In (ii), at $\phi \to 0$ the curve starts at $(1,0)$ on the unit circle, and the normal there tends to the vertical line $x = 1$, which is indeed a tangent to $x^2 + y^2 = 1$.
- A computer check over a thousand random values of ϕ confirmed all the normal equations, the tangency in (i) and the unit distance in (ii) to within rounding.

Common slips

- Getting $\mathrm{d}y/\mathrm{d}t$ wrong in (i): $12\cos t - 12\sin^2 t\cos t$ simplifies to $12\cos^3 t$, and the factorisation is what makes the gradient so simple.
- Only checking that P lies on the normal in (i). A line through a point of a curve is not automatically a tangent there; the gradients must match too.
- Answering "a circle" in (ii) without giving its equation, centre and radius.

Beyond STEP

The envelope of all the normals to a curve is called its *evolute*. It is the locus of the centres of curvature. So part (i) shows that the astroid $x^{2/3} + y^{2/3} = 4$ is the evolute of the first curve. In (ii) the curve is the *involute* of the unit circle (the path traced by the end of a taut string unwound from a reel), and its evolute is the circle. Involutes of circles give the profile of most gear teeth. These ideas belong to the differential geometry of curves. Needham, *Visual Complex Analysis*, is a beautifully geometric companion to this kind of reasoning about plane curves.

WORKED SOLUTION

STEP III 2021, Question 2

Pure • Determinants and Inequalities

STEP III 2021, Question 2

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

The key idea

Each definition of x, y, z can be cleared of fractions to give an equation that is *linear* in a, b, c. Three such equations make a homogeneous linear system with a non-zero solution (a, b, c), so the matrix of coefficients must be singular: its determinant is zero. That determinant is exactly the symmetric expression in x, y, z we want. After that, the inequalities come from rewriting the target in terms of x, y, z and using basic facts about sums of squares and AM–GM.

Solution

Part (i)

Since a, b, c are distinct, none of the denominators is zero. Clearing fractions,

$$x(b-c) = a, \qquad y(c-a) = b, \qquad z(a-b) = c.$$

Now multiply out the matrix product row by row:

$$\begin{aligned} a - xb + xc &= a - x(b-c) = a - a = 0, \\ ya + b - yc &= b - y(c-a) = b - b = 0, \\ -za + zb + c &= c - z(a-b) = c - c = 0, \end{aligned}$$

which is the required matrix equation.

Call the 3×3 matrix M. Because a, b, c are distinct, they are not all zero, so M sends a *non-zero* vector to the zero vector.

Ask yourself Why does this force $\det M = 0$?

If $\det M \neq 0$ then M^{-1} exists, and multiplying $M\mathbf{v} = \mathbf{0}$ on the left by M^{-1} would give

$\mathbf{v} = \mathbf{0}$, contradicting $(a, b, c) \neq (0, 0, 0)$. So $\det M = 0$. Expanding along the first row,

$$\begin{aligned}\det M &= 1 \cdot (1 + yz) + x(y - yz) + x(yz + z) \\ &= 1 + yz + xy - xyz + xyz + xz \\ &= 1 + xy + yz + zx.\end{aligned}$$

Setting this equal to zero gives $yz + zx + xy = -1$.

For the inequality, the left-hand side is $x^2 + y^2 + z^2$. Using the identity $(x + y + z)^2 = x^2 + y^2 + z^2 + 2(xy + yz + zx)$,

$$x^2 + y^2 + z^2 = (x + y + z)^2 - 2(xy + yz + zx) = (x + y + z)^2 + 2 \geq 2,$$

since the square of the real number $x + y + z$ is non-negative. This is the required result.

Part (ii)

Here the denominators are positive, and clearing fractions gives $x(b + c) = 2a$ and so on. Rearranged as linear equations in a, b, c:

$$\begin{pmatrix} -2 & x & x \\ y & -2 & y \\ z & z & -2 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

(The first row reads $-2a + x(b + c) = 0$, and the others are similar.) As $a, b, c > 0$ the vector is non-zero, so, exactly as in part (i), the determinant vanishes:

$$\begin{aligned}0 &= -2(4 - yz) - x(-2y - yz) + x(yz + 2z) \\ &= -8 + 2yz + 2xy + xyz + xyz + 2xz \\ &= -8 + 2(xy + yz + zx) + 2xyz.\end{aligned}$$

Dividing by 2 gives $xyz + yz + zx + xy = 4$.

The first inequality. The trick is to divide each bracket on the left by the matching bracket on the right:

$$\frac{2a + b + c}{b + c} = \frac{2a}{b + c} + 1 = 1 + x,$$

and similarly for the other two. Since $(b + c)(c + a)(a + b) > 0$, the inequality is equivalent to $(1 + x)(1 + y)(1 + z) > 5$. Expanding and using the result just proved,

$$(1 + x)(1 + y)(1 + z) = 1 + (x + y + z) + (xy + yz + zx) + xyz = 5 + (x + y + z).$$

As x, y, z are all positive, this exceeds 5, which proves the first inequality.

The second inequality. By the same calculation we now need $x + y + z > 2$. Write $S = x + y + z$.

Ask yourself We know $xy + yz + zx + xyz = 4$. How can that force S to be reasonably large?

Both $xy + yz + zx$ and xyz are bounded *above* in terms of S:

- $S^2 - 3(xy + yz + zx) = \frac{1}{2}\left[(x-y)^2 + (y-z)^2 + (z-x)^2\right] \geq 0$, so $xy + yz + zx \leq \frac{1}{3}S^2$;
- by AM–GM for the positive numbers x, y, z, we have $xyz \leq \left(\frac{1}{3}S\right)^3 = \frac{1}{27}S^3$.

Adding,

$$4 = xy + yz + zx + xyz \leq \tfrac{1}{3}S^2 + \tfrac{1}{27}S^3.$$

The right-hand side is an increasing function of S for $S > 0$, and it equals $3 + 1 = 4$ when $S = 3$. So if $S < 3$ the right-hand side would be less than 4, which is impossible. Hence $S \geq 3$, and

$$\frac{(2a+b+c)(a+2b+c)(a+b+2c)}{(b+c)(c+a)(a+b)} = 5 + S \geq 8 > 7,$$

which proves the second inequality (and in fact gives the sharper bound 8, attained when $a = b = c$).

Checks

- Take $a = b = c = 1$ in part (ii): the left-hand side is $4^3 = 64$ and $(b+c)(c+a)(a+b) = 8$, so the ratio is 8, matching $5 + S$ with $x = y = z = 1$, $S = 3$. Also $xyz + xy + yz + zx = 1 + 3 = 4$.
- In part (i), take $(a, b, c) = (0, 1, 2)$: then $x = 0$, $y = \frac{1}{2}$, $z = -2$, and $xy + yz + zx = -1$ as required; the sum of squares is $\frac{17}{4} \geq 2$.
- A computer check over 20 000 random rational triples confirmed both identities exactly; the smallest ratio found in part (ii) was about 8.0001, and for $a = b = 1$, $c = 10^{-6}$ it is about 9.0, so the bound 8 is the true minimum, comfortably above 7.

Common slips

- Asserting $\det M = 0$ without saying why: the key point is that (a, b, c) is a *non-zero* vector, which needs "distinct" in part (i) and "positive" in part (ii).
- Dividing $2a + b + c$ by $b + c$ and getting $1 + \frac{1}{2}x$ instead of $1 + x$: remember x already contains the factor 2.
- Using $xy + yz + zx < 4$ (an *upper* bound) to try to bound S from below; the inequalities have to point the other way, as above.

Beyond STEP

The fact that a square matrix kills a non-zero vector exactly when its determinant is zero is the heart of linear algebra: it is how eigenvalues are found, via $\det(A - \lambda I) = 0$. The bound $S \geq 3$ in part (ii) is Nesbitt's inequality, $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}$, in disguise. For the linear algebra, see Strang, *Introduction to Linear Algebra*; for more inequalities of this kind, Siklos, *Advanced Problems in Mathematics*, is excellent practice.

WORKED SOLUTION

STEP III 2021, Question 3

Pure • Integral Inequalities

STEP III 2021, Question 3

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

The key idea

Differentiate the thing being raised to the power n. For $u = \sec x + \tan x$ the derivative is $u \sec x$, which makes $u^{n-1}\frac{du}{dx}$ exactly integrable, so the job is to rewrite the integrand of $I_{n+1} + I_{n-1}$ in that form. The inequalities then come from comparing the integrand $u^n = u^{n-1} \cdot u$ with the exactly integrable $u^{n-1} \cdot u'$: if $u < u'$, the integral is less than $\left[u^n/n\right]$.

Solution

Part (i)

Let $u = \sec x + \tan x$. Then

$$\frac{du}{dx} = \sec x \tan x + \sec^2 x = \sec x\,(\sec x + \tan x) = u \sec x.$$

Also, using $\tan^2 x + 1 = \sec^2 x$,

$$u^2 + 1 = \sec^2 x + 2\sec x \tan x + \tan^2 x + 1 = 2\sec^2 x + 2\sec x \tan x = 2u \sec x = 2\frac{du}{dx}.$$

So for $n \geq 1$,

$$\begin{aligned} I_{n+1} + I_{n-1} &= \int_0^\beta u^{n-1}\left(u^2+1\right)\mathrm{d}x = 2\int_0^\beta u^{n-1}\frac{du}{dx}\,\mathrm{d}x = \frac{2}{n}\left[u^n\right]_0^\beta \\ &= \frac{2}{n}\left((\sec\beta + \tan\beta)^n - 1\right), \end{aligned}$$

since $u = 1$ at $x = 0$. Halving gives the required identity.

Ask yourself The identity involves I_{n+1} and I_{n-1}, but the inequality is about I_n. How can the middle term be squeezed in?

Compare u^2+1 with $2u$. We have $u^2+1-2u=(u-1)^2\geq 0$, and for $0<x\leq\beta$ we have $\sec x>1$ and $\tan x>0$, so $u>1$ and the inequality is strict there. Multiplying by $u^{n-1}>0$ gives $u^{n+1}+u^{n-1}>2u^n$ on $(0,\beta]$, and integrating (the integrands are continuous and the inequality is strict on an interval of positive length) gives

$$I_{n+1}+I_{n-1}>2I_n, \qquad \text{so} \qquad I_n<\tfrac{1}{2}\left(I_{n+1}+I_{n-1}\right)=\frac{1}{n}\left((\sec\beta+\tan\beta)^n-1\right).$$

(Equivalently: $u'-u=u(\sec x-1)>0$ for $x>0$, so $I_n=\int_0^\beta u^{n-1}u\,\mathrm{d}x<\int_0^\beta u^{n-1}u'\,\mathrm{d}x$. This view is the one that carries over to part (ii).)

Part (ii)

Write $c=\cos\beta$, so $0<c<1$, and let $v=c\sec x+\tan x$. Then

$$v'=\frac{\mathrm{d}v}{\mathrm{d}x}=c\sec x\tan x+\sec^2 x.$$

At the endpoints, $v(0)=c=\cos\beta$ and $v(\beta)=1+\tan\beta$, which are exactly the two numbers in the target. So the bound is $\frac{1}{n}\left[v^n\right]_0^\beta=\int_0^\beta v^{n-1}v'\,\mathrm{d}x$, and it is enough to show that $v<v'$ throughout $[0,\beta]$.

> **Ask yourself** In part (i), u^2+1 was exactly $2u'$. What is the corresponding relation for v?

Repeating the calculation of part (i),

$$v^2+1=c^2\sec^2 x+2c\sec x\tan x+\tan^2 x+1=(1+c^2)\sec^2 x+2c\sec x\tan x,$$

so

$$2v'-(v^2+1)=(1-c^2)\sec^2 x=\sin^2\beta\,\sec^2 x.$$

It is no longer an equality, but the leftover term is positive. Combining with $v^2+1-2v=(v-1)^2$,

$$2(v'-v)=(v-1)^2+\sin^2\beta\,\sec^2 x>0,$$

because $\sin\beta\neq 0$ for $0<\beta<\frac{\pi}{2}$. Hence $v<v'$ for every $x\in[0,\beta]$.

Finally $v>0$ on $[0,\beta]$ (as $c>0$, $\sec x>0$ and $\tan x\geq 0$), so $v^{n-1}>0$ and $v^n=v^{n-1}v<v^{n-1}v'$. Integrating over $[0,\beta]$, for $n\geq 1$,

$$J_n=\int_0^\beta v^n\,\mathrm{d}x<\int_0^\beta v^{n-1}\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x=\frac{1}{n}\left[v^n\right]_0^\beta=\frac{1}{n}\left((1+\tan\beta)^n-\cos^n\beta\right),$$

as required.

Checks

- With $n=1$ the identity in (i) says $\frac{1}{2}(I_2+\beta)=\sec\beta+\tan\beta-1$. For $\beta=1$ a numerical integration gives both sides as $2.40822\ldots$.

- For $\beta = 1$, $n = 3$: numerically $I_3 \approx 8.774$ against the bound 12.863, and $J_3 \approx 3.326$ against the bound 5.523.
- Putting $\cos\beta = 1$ formally in (ii) turns J_n into I_n and the leftover term $\sin^2\beta \sec^2 x$ disappears: part (ii) really is a perturbation of part (i).

Common slips

- Forgetting to check positivity: multiplying $v < v'$ by v^{n-1} needs $v^{n-1} > 0$, which is why $v > 0$ matters.
- Claiming only $\leq$: the strict inequality needs $u > 1$ (or $\sin\beta \neq 0$) on an interval, not just at a point.
- Evaluating $v(\beta)$ carelessly: $\sec\beta\cos\beta = 1$, which is where the $1 + \tan\beta$ comes from.

Beyond STEP

The argument "if $0 < f \leq g$ pointwise, with strict inequality somewhere, then $\int f < \int g$" is the monotonicity of the integral, which an analysis course proves carefully from the definition of the Riemann integral. Comparing an awkward integrand with an exact derivative is a standard way to bound integrals and sums, and it reappears in estimates such as the integral test. Abbott's *Understanding Analysis* covers integration and these comparison arguments rigorously.

WORKED SOLUTION

STEP III 2021, Question 4

Pure • Vectors and Projections

STEP III 2021, Question 4

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

The key idea

Everything here is done with the scalar product: an angle between two vectors is read off from $\mathbf{u}\cdot\mathbf{v} = |\mathbf{u}||\mathbf{v}|\cos(\text{angle})$, and projection is linear, so the projection of $\frac{1}{2}(\mathbf{a}+\mathbf{b})$ is $\frac{1}{2}(\mathbf{a}_1+\mathbf{b}_1)$. Part (i) shows that the average of two *equal-length* vectors bisects the angle between them. In part (iii) the projections $\mathbf{a}_1$ and $\mathbf{b}_1$ usually have different lengths, so we should expect bisection to fail unless something special happens. The condition factorises neatly into the two special cases.

Solution

Part (i)

Since $|\mathbf{a}| = |\mathbf{b}| = 1$ and $\mathbf{a}\cdot\mathbf{b} = \cos\theta$,

$$|\mathbf{m}|^2 = \tfrac{1}{4}\left(1 + 2\cos\theta + 1\right) = \tfrac{1}{2}(1+\cos\theta) = \cos^2\tfrac{\theta}{2},$$

so $|\mathbf{m}| = \cos\frac{\theta}{2}$, which is positive because $0 < \theta < \pi$. In particular $\mathbf{m} \neq \mathbf{0}$. Also

$$\mathbf{m}\cdot\mathbf{a} = \tfrac{1}{2}(1+\cos\theta) = \cos^2\tfrac{\theta}{2} = \mathbf{m}\cdot\mathbf{b}.$$

So the cosine of the angle between $\mathbf{m}$ and $\mathbf{a}$ is $\cos^2\frac{\theta}{2}\big/\cos\frac{\theta}{2} = \cos\frac{\theta}{2}$, and the same holds for $\mathbf{b}$. Both angles are therefore $\frac{\theta}{2}$. Since $\mathbf{m}$ lies in the plane of $\mathbf{a}$ and $\mathbf{b}$ and the two angles add up to θ, $\mathbf{m}$ bisects the angle between $\mathbf{a}$ and $\mathbf{b}$.

Part (ii)

By definition, with $\mathbf{c}$ as the unit normal and $\mathbf{a}\cdot\mathbf{c} = \cos\alpha$,

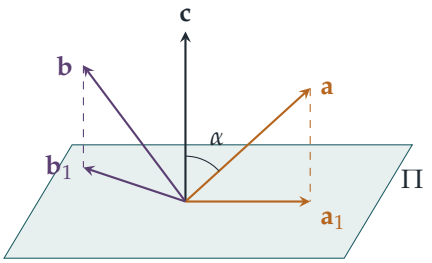
$$\mathbf{a}_1 = \mathbf{a} - \cos\alpha\,\mathbf{c}, \qquad \mathbf{b}_1 = \mathbf{b} - \cos\beta\,\mathbf{c}.$$

Then $\mathbf{a}_1\cdot\mathbf{c} = \cos\alpha - \cos\alpha\,(\mathbf{c}\cdot\mathbf{c}) = 0$, since $|\mathbf{c}| = 1$. Similarly $\mathbf{b}_1\cdot\mathbf{c} = 0$.

Because $\mathbf{a}_1 \cdot \mathbf{c} = 0$, we get $|\mathbf{a}_1|^2 = \mathbf{a}_1 \cdot (\mathbf{a} - \cos\alpha\,\mathbf{c}) = \mathbf{a}_1 \cdot \mathbf{a}$, and

$$\mathbf{a}_1 \cdot \mathbf{a} = \mathbf{a} \cdot \mathbf{a} - \cos\alpha\,(\mathbf{c} \cdot \mathbf{a}) = 1 - \cos^2\alpha = \sin^2\alpha.$$

As α is acute, $\sin\alpha > 0$, so $|\mathbf{a}_1| = \sin\alpha$. In the same way $|\mathbf{b}_1| = \sin\beta$. Both are nonzero, so the angle ϕ is defined.



For the angle between the projections, again use $\mathbf{a}_1 \cdot \mathbf{c} = 0$ to drop a term:

$$\mathbf{a}_1 \cdot \mathbf{b}_1 = \mathbf{a}_1 \cdot (\mathbf{b} - \cos\beta\,\mathbf{c}) = \mathbf{a}_1 \cdot \mathbf{b} = \mathbf{a} \cdot \mathbf{b} - \cos\alpha\,(\mathbf{c} \cdot \mathbf{b}) = \cos\theta - \cos\alpha\cos\beta.$$

Dividing by $|\mathbf{a}_1||\mathbf{b}_1| = \sin\alpha\sin\beta$ gives

$$\cos\phi = \frac{\cos\theta - \cos\alpha\cos\beta}{\sin\alpha\sin\beta},$$

as required.

Part (iii)

Projection is linear, so

$$\mathbf{m}_1 = \mathbf{m} - (\mathbf{m} \cdot \mathbf{c})\,\mathbf{c} = \tfrac{1}{2}\big(\mathbf{a} - \cos\alpha\,\mathbf{c}\big) + \tfrac{1}{2}\big(\mathbf{b} - \cos\beta\,\mathbf{c}\big) = \tfrac{1}{2}(\mathbf{a}_1 + \mathbf{b}_1).$$

We assume $\mathbf{m}_1 \neq \mathbf{0}$, since otherwise it has no direction and cannot bisect anything. As $\mathbf{m}_1$ is a positive combination of $\mathbf{a}_1$ and $\mathbf{b}_1$, it lies in their plane and between them. So it bisects the angle exactly when it makes equal angles with $\mathbf{a}_1$ and $\mathbf{b}_1$:

$$\frac{\mathbf{m}_1 \cdot \mathbf{a}_1}{|\mathbf{a}_1|} = \frac{\mathbf{m}_1 \cdot \mathbf{b}_1}{|\mathbf{b}_1|}.$$

(We have cancelled the common factor $|\mathbf{m}_1|$, and cos is one-to-one on $[0, \pi]$.)

? **Ask yourself** Why not just apply part (i) to $\mathbf{a}_1$ and $\mathbf{b}_1$?

Part (i) needed $|\mathbf{a}| = |\mathbf{b}|$. Here $|\mathbf{a}_1| = \sin\alpha$ and $|\mathbf{b}_1| = \sin\beta$, which are usually different. So we must compute.

Write $P = \mathbf{a}_1 \cdot \mathbf{b}_1$, $s = \sin\alpha$ and $t = \sin\beta$. Then

$$\mathbf{m}_1 \cdot \mathbf{a}_1 = \tfrac{1}{2}(s^2 + P), \qquad \mathbf{m}_1 \cdot \mathbf{b}_1 = \tfrac{1}{2}(P + t^2).$$

Multiplying through by $2st > 0$, the condition becomes

$$\begin{aligned} t(s^2+P) = s(t^2+P) &\iff st(s-t) - P(s-t) = 0 \\ &\iff (s-t)(st-P) = 0. \end{aligned}$$

So $\mathbf{m}_1$ bisects the angle if and only if $s = t$ or $P = st$.

- $s = t$ means $\sin\alpha = \sin\beta$. Since sin is strictly increasing on $(0, \frac{\pi}{2})$ and both angles are acute, this holds if and only if $\alpha = \beta$.
- By part (ii), $P = \cos\theta - \cos\alpha\cos\beta$. So $P = st$ means

$$\cos\theta = \cos\alpha\cos\beta + \sin\alpha\sin\beta = \cos(\alpha - \beta).$$

Each step was an equivalence, so $\mathbf{m}_1$ bisects the angle between $\mathbf{a}_1$ and $\mathbf{b}_1$ if and only if $\alpha = \beta$ or $\cos\theta = \cos(\alpha - \beta)$.

Geometrically, the first case says $\mathbf{a}_1$ and $\mathbf{b}_1$ have equal length, so part (i) applies after all. The second says $\cos\phi = 1$: the projections point the same way (when $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are coplanar with $\mathbf{a}$ and $\mathbf{b}$ on the same side of $\mathbf{c}$). The angle is then zero, and it is bisected trivially.

Checks

- Take $\alpha = \beta = \frac{\pi}{2}$ (not allowed here, but a useful limit). Then $\mathbf{a}$ and $\mathbf{b}$ already lie in the plane, and the formula gives $\cos\phi = \cos\theta$, as it should.
- With $\alpha = \frac{\pi}{3}$, $\beta = \frac{\pi}{4}$ and $\theta = \frac{\pi}{2}$, the formula gives $\cos\phi = -\frac{\sqrt{2}/4}{\sqrt{6}/4} = -1/\sqrt{3} \approx -0.577$. This agrees with constructing the vectors explicitly.
- Both the formula for $\cos\phi$ and the factorised bisection condition were confirmed by computer on several thousand random triples of unit vectors.

Common slips

- Forgetting that $|\mathbf{a}_1| \neq 1$. The projection of a unit vector is not a unit vector, so the denominator $\sin\alpha\sin\beta$ is essential.
- Assuming the bisector of $\mathbf{a}_1, \mathbf{b}_1$ is always $\mathbf{a}_1 + \mathbf{b}_1$. That holds only for equal lengths. In general the bisector is $\mathbf{a}_1/|\mathbf{a}_1| + \mathbf{b}_1/|\mathbf{b}_1|$.
- Dividing by $\sin\alpha - \sin\beta$ instead of factorising, and so losing the case $\alpha = \beta$.

Beyond STEP

Projection onto a plane is a *linear map* $\mathbf{x} \mapsto \mathbf{x} - (\mathbf{x}\cdot\mathbf{n})\mathbf{n}$, with matrix $I - \mathbf{n}\mathbf{n}^{\mathsf{T}}$. Linear maps preserve sums but not lengths or angles, and that is why this question works out as it does. The formula for $\cos\phi$ is the spherical cosine rule in disguise. Here ϕ is the

angle at the vertex **c** of the spherical triangle with vertices **a**, **b** and **c**, whose sides are α, β and θ. For projections and orthogonality, see Strang, *Introduction to Linear Algebra*.

WORKED SOLUTION

STEP III 2021, Question 5

Pure • Polar Curves

STEP III 2021, Question 5

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

The key idea

Both curves are given as r in terms of θ, so they meet where the two expressions for r agree, and they touch where, in addition, the two values of $\mathrm{d}r/\mathrm{d}\theta$ agree. Writing everything in terms of $c = \cos\theta$ turns the meeting condition into a quadratic in c. A repeated root of that quadratic gives one way of touching, but it isn't the only way: because $\cos\theta$ is itself stationary at $\theta = 0$ and $\theta = \pi$, a simple root at $c = \pm 1$ also gives a double zero in θ.

Solution

Throughout, write $c = \cos\theta$, and

$$f(\theta) = a + 2\cos\theta, \qquad g(\theta) = 2 + \cos 2\theta = 1 + 2\cos^2\theta.$$

Note that $g(\theta) \geq 1$ for all θ, so the second curve never passes through the origin O. Away from O, a point with $r > 0$ has a unique angle θ (up to multiples of 2π). So the curves can only meet at points where they share both r and θ.

Part (i)

The curves meet when $f(\theta) = g(\theta)$, that is $a + 2c = 1 + 2c^2$, or

$$2\cos^2\theta - 2\cos\theta + 1 - a = 0,$$

as required. It helps to name the difference:

$$h(\theta) = f(\theta) - g(\theta) = -(2c^2 - 2c + 1 - a).$$

When do they touch? At a common point, the angle ψ between the radius and the tangent satisfies $\tan\psi = r \big/ \frac{\mathrm{d}r}{\mathrm{d}\theta}$. So two curves through the same point (r, θ) have the same tangent there exactly when they also have the same $\mathrm{d}r/\mathrm{d}\theta$. The curves therefore touch where $h(\theta) = 0$ and $h'(\theta) = 0$. Differentiating,

$$h'(\theta) = -2\sin\theta + 2\sin 2\theta = 2\sin\theta\,(2\cos\theta - 1).$$

So $h'(\theta) = 0$ exactly when $\cos\theta = \frac{1}{2}$ or $\sin\theta = 0$.

Ask yourself The quadratic in c has a repeated root only when $a = \frac{1}{2}$. So why are there other values of a?

A repeated root in c is enough for touching, but it isn't necessary. The map $\theta \mapsto \cos\theta$ is stationary at $\theta = 0$ and $\theta = \pi$. So if $c = \pm 1$ is a root at all, h has a double zero in θ there. The condition $h' = 0$ captures both possibilities.

- $\cos\theta = \frac{1}{2}$: substituting into the meeting equation gives $\frac{1}{2} - 1 + 1 - a = 0$, so $a = \frac{1}{2}$. (Equivalently, the discriminant $4 - 8(1 - a)$ vanishes when $a = \frac{1}{2}$, and then the repeated root is $c = \frac{1}{2}$.) So the curves touch when $a = \frac{1}{2}$, as required.
- $\theta = 0$, $c = 1$: we need $2 - 2 + 1 - a = 0$, so $a = 1$. Then $r = 3$ on both curves.
- $\theta = \pi$, $c = -1$: we need $2 + 2 + 1 - a = 0$, so $a = 5$. Then $r = 5 - 2 = 3$ on both curves, so $r \geq 0$ holds.

Hence the other two values are $a = 1$ and $a = 5$.

Features of the second curve

All three sketches contain $r = 2 + \cos 2\theta$, so we describe it once. It is unchanged by $\theta \to -\theta$ and by $\theta \to \pi - \theta$, so it is symmetric in both axes. Here r goes from a maximum of 3 at $\theta = 0, \pi$ to a minimum of 1 at $\theta = \pm\frac{1}{2}\pi$. It is *not* an ellipse. Near $\theta = \frac{1}{2}\pi$, with $s = \sin\theta$,

$$y = r\sin\theta = (3 - 2s^2)s = 3s - 2s^3, \qquad \frac{\mathrm{d}y}{\mathrm{d}s} = 3 - 6s^2 = -3 \text{ at } s = 1.$$

So as θ moves away from $\frac{1}{2}\pi$ (and s falls below 1), y *increases*. The curve has a smooth indentation (a dimple) at $(0, \pm 1)$, not a cusp, since $\mathrm{d}r/\mathrm{d}\theta = -2\sin 2\theta = 0$ there and the tangent is horizontal. By contrast, $x = r\cos\theta = c + 2c^3$ is increasing in c, so the ends at $(\pm 3, 0)$ are simply rounded.

The first curve $r = a + 2\cos\theta$ is symmetric about the initial line, since it is unchanged by $\theta \to -\theta$. It has $r = a + 2$ at $\theta = 0$, $r = a$ at $\theta = \pm\frac{1}{2}\pi$ and (when $a \geq 2$) $r = a - 2$ at $\theta = \pi$. When $a < 2$, the restriction $r \geq 0$ cuts it off where $\cos\theta = -\frac{1}{2}a$, and it reaches O along those two directions.

Part (ii)

With $a = \frac{1}{2}$,

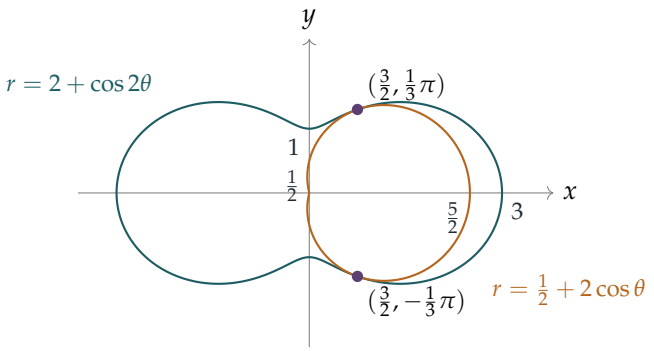
$$h(\theta) = -\left(2c^2 - 2c + \tfrac{1}{2}\right) = -2\left(c - \tfrac{1}{2}\right)^2 \leq 0.$$

So the first curve lies inside the second everywhere and meets it only where $\cos\theta = \frac{1}{2}$. They touch at

$$\theta = \pm\tfrac{1}{3}\pi, \qquad r = \tfrac{1}{2} + 1 = \tfrac{3}{2}.$$

(Check: $2 + \cos\frac{2}{3}\pi = \frac{3}{2}$, and both values of $\mathrm{d}r/\mathrm{d}\theta$ equal $-\sqrt{3}$ at $\theta = \frac{1}{3}\pi$.)

The first curve exists only for $\cos\theta \geq -\frac{1}{4}$, that is $|\theta| \leq \arccos(-\frac{1}{4}) \approx 104.5°$. It runs from $r = \frac{5}{2}$ on the initial line, through $r = \frac{1}{2}$ at $\theta = \pm\frac{1}{2}\pi$, down to O. It arrives at O along the lines $\theta = \pm\arccos(-\frac{1}{4})$, so it has a corner (a pointed tip) at the origin, just to the left of the y-axis.

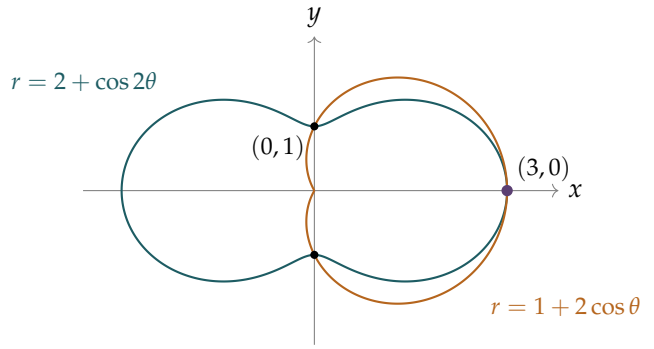


Part (iii)

The case $a = 1$. Here $h(\theta) = -(2c^2 - 2c) = 2c(1 - c)$. The curves meet where $c = 1$ or $c = 0$.

- They touch at $\theta = 0$, $r = 3$, the point $(3, 0)$. Near $\theta = 0$ we have $0 < c < 1$, so $h > 0$ and the first curve is just outside the second on both sides of the point of contact.
- At $\theta = \pm\frac{1}{2}\pi$, $r = 1$ on both curves: the points $(0, \pm 1)$, the bottoms of the dimples. Here h changes sign with c, so the curves *cross*. The first curve is outside for $|\theta| < \frac{1}{2}\pi$ and inside for $|\theta| > \frac{1}{2}\pi$.

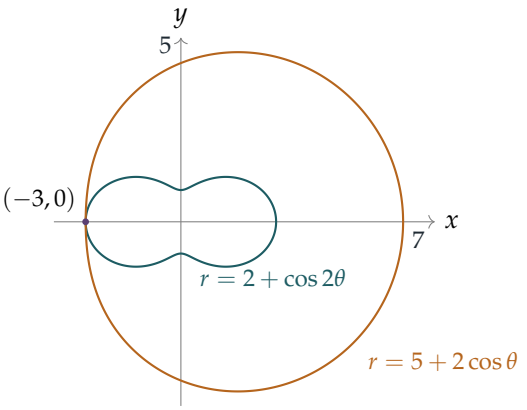
The first curve exists for $\cos\theta \geq -\frac{1}{2}$, i.e. $|\theta| \leq \frac{2}{3}\pi$. It reaches O along the lines $\theta = \pm\frac{2}{3}\pi$, making a corner there.



The case $a = 5$. Here $h(\theta) = -(2c^2 - 2c - 4) = 2(2 - c)(1 + c) \geq 0$, with equality only at $c = -1$. So the first curve lies entirely outside the second and touches it only at

$$\theta = \pi, \qquad r = 3,$$

the point $(-3, 0)$. Since $a > 2$, now $r > 0$ throughout, so the first curve is a closed loop (a limaçon with no dimple, because $a \geq 4$). It has $r = 7$ at $\theta = 0$, $r = 5$ at $\theta = \pm\frac{1}{2}\pi$ and $r = 3$ at $\theta = \pi$.



Checks

- At each touching point, compare both values of r and of $\mathrm{d}r/\mathrm{d}\theta$. For $a = 1$ at $\theta = 0$, both $r = 3$ and both derivatives are 0; the same holds for $a = 5$ at $\theta = \pi$.
- A computer scan over $-3 \leq a < 9$ in steps of 0.01 found $h = h' = 0$ (with $r \geq 0$) only for $a = \frac{1}{2}$, 1 and 5. This matches the factorisation $h' = 2\sin\theta(2\cos\theta - 1)$.
- The point $(0, 1)$ lies on the second curve, and a nearby point at $\theta = \frac{1}{2}\pi + 0.1$ has $y \approx 1.015 > 1$. This confirms the dimple.

Common slips

- Stopping at the discriminant and missing $a = 1$ and $a = 5$. A simple root in $c = \cos\theta$ can still be a double root in θ.
- Drawing $r = 2 + \cos 2\theta$ as an ellipse, or giving it cusps at $(0, \pm 1)$ instead of smooth dimples.
- Drawing the inner loop of the limaçon for $a = \frac{1}{2}$ or $a = 1$. That loop has $r < 0$, which the question excludes.

Beyond STEP

Touching is really a statement about multiplicity. Two curves are tangent where the difference of their equations has a repeated zero along the curve, and the parametrisation can create or hide such zeros, as $\cos\theta$ did here. In a first course on curves, this becomes the idea of the order of contact, and the curvature $\left(r^2 + 2r'^2 - rr''\right) / \left(r^2 + r'^2\right)^{3/2}$ of a polar curve explains exactly where the dimples of $2 + \cos 2\theta$ come from. For more on

parametrised curves and their geometry, try Siklos, *Advanced Problems in Mathematics*, and for the analysis behind smoothness and tangency, Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP III 2021, Question 6

Pure • Inverse Functions and Derivatives

STEP III 2021, Question 6

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

The key idea

Both functions look forbidding, but their derivatives turn out to be very simple. A function whose derivative is $\frac{1}{1+x^2}$ must be $\tan^{-1} x$ plus a constant, and a function with zero derivative is constant, but only on each interval where it is defined. So the real work is to find the constant on each separate piece, using a convenient value or a limit, and to watch the jumps at the points where the function is undefined.

Solution

Part (i)

Write $t = \tan\alpha$, so $t > 0$, and let $u = \dfrac{xt+1}{t-x}$, so $f_\alpha(x) = \tan^{-1} u$. By the quotient rule,

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{t(t-x)+(xt+1)}{(t-x)^2} = \frac{1+t^2}{(t-x)^2},$$

and

$$1+u^2 = \frac{(t-x)^2+(xt+1)^2}{(t-x)^2} = \frac{t^2+x^2+x^2t^2+1}{(t-x)^2} = \frac{(1+t^2)(1+x^2)}{(t-x)^2}.$$

So by the chain rule

$$f'_\alpha(x) = \frac{1}{1+u^2}\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{1+x^2},$$

as required.

? **Ask yourself** Does this mean $f_\alpha(x) = \tan^{-1} x + c$ for a single constant c?

No. The function is undefined at $x = \tan\alpha$, so the domain is two separate intervals, and on each one $f_\alpha(x) - \tan^{-1} x$ has zero derivative and so is constant, but the two constants need not agree.

For $x < \tan\alpha$: at $x = 0$ we have $f_\alpha(0) = \tan^{-1}(\cot\alpha) = \frac{1}{2}\pi - \alpha$, since $0 < \frac{1}{2}\pi - \alpha < \frac{1}{2}\pi$. Hence

$$f_\alpha(x) = \tan^{-1} x + \tfrac{1}{2}\pi - \alpha \qquad (x < \tan\alpha).$$

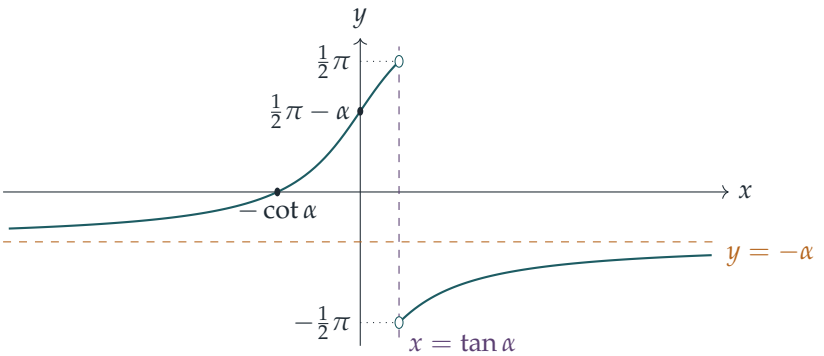
For $x > \tan\alpha$: as $x \to \tan\alpha$ from above, the numerator $xt + 1$ tends to $1 + t^2 > 0$ while the denominator $t - x$ tends to 0 through negative values, so $u \to -\infty$ and $f_\alpha(x) \to -\frac{1}{2}\pi$. Since $\tan^{-1} x \to \alpha$, the constant is $-\frac{1}{2}\pi - \alpha$:

$$f_\alpha(x) = \tan^{-1} x - \tfrac{1}{2}\pi - \alpha \qquad (x > \tan\alpha).$$

(The same limit argument from below gives $u \to +\infty$ and the value $\frac{1}{2}\pi$, agreeing with the first formula.)

So the graph is the graph of $y = \tan^{-1} x$ in two pieces, one shifted up by $\frac{1}{2}\pi - \alpha$ and the other shifted down by $\frac{1}{2}\pi + \alpha$. Its features are:

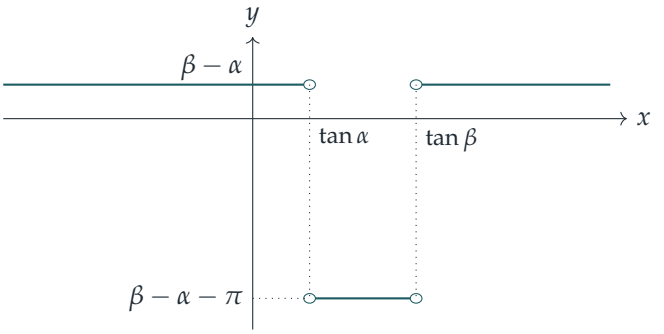
- it is increasing on each piece, with gradient $\frac{1}{1+x^2}$;
- as $x \to \pm\infty$, $y \to -\alpha$ (from above on the left, from below on the right), so $y = -\alpha$ is an asymptote at both ends;
- the left piece runs from $-\alpha$ up to $\frac{1}{2}\pi$ (not attained) at $x = \tan\alpha$, crossing the y-axis at $\frac{1}{2}\pi - \alpha$ and the x-axis where $\tan^{-1} x = \alpha - \frac{1}{2}\pi$, that is at $x = -\cot\alpha$;
- the right piece starts just above $-\frac{1}{2}\pi$ at $x = \tan\alpha$ and rises to $-\alpha$; it lies entirely below the x-axis;
- at $x = \tan\alpha$ there is a jump of $-\pi$, and both pieces have gradient $\cos^2\alpha$ there.



The difference. Since $0 < \alpha < \beta < \frac{1}{2}\pi$ we have $\tan\alpha < \tan\beta$, so there are three intervals. Using the two formulas for each function:

- for $x < \tan\alpha$: $f_\alpha(x) - f_\beta(x) = (\frac{1}{2}\pi - \alpha) - (\frac{1}{2}\pi - \beta) = \beta - \alpha$;
- for $\tan\alpha < x < \tan\beta$: $f_\alpha(x) - f_\beta(x) = (-\frac{1}{2}\pi - \alpha) - (\frac{1}{2}\pi - \beta) = \beta - \alpha - \pi$;
- for $x > \tan\beta$: $f_\alpha(x) - f_\beta(x) = (-\frac{1}{2}\pi - \alpha) - (-\frac{1}{2}\pi - \beta) = \beta - \alpha$.

The $\tan^{-1} x$ terms cancel, so the graph is a step function: the horizontal line $y = \beta - \alpha$ (which lies between 0 and $\frac{1}{2}\pi$), except on the interval between $\tan\alpha$ and $\tan\beta$, where it drops by π to $y = \beta - \alpha - \pi$ (which lies between $-\pi$ and $-\frac{1}{2}\pi$). The function is undefined at $x = \tan\alpha$ and $x = \tan\beta$.



Part (ii)

Differentiate each term. Since $\dfrac{\mathrm{d}}{\mathrm{d}s}\tanh^{-1} s = \dfrac{1}{1-s^2}$,

$$\frac{\mathrm{d}}{\mathrm{d}x}\tanh^{-1}(\sin x) = \frac{\cos x}{1-\sin^2 x} = \frac{1}{\cos x} = \sec x,$$

and since $\dfrac{\mathrm{d}}{\mathrm{d}s}\sinh^{-1} s = \dfrac{1}{\sqrt{1+s^2}}$,

$$\frac{\mathrm{d}}{\mathrm{d}x}\sinh^{-1}(\tan x) = \frac{\sec^2 x}{\sqrt{1+\tan^2 x}} = \frac{\sec^2 x}{\sqrt{\sec^2 x}} = \frac{\sec^2 x}{|\sec x|} = |\sec x|.$$

The square root is the positive one, which is why the modulus appears. So $g'(x) = \sec x - |\sec x|$. For $\frac{1}{2}\pi < x < \frac{3}{2}\pi$ we have $\cos x < 0$, so $|\sec x| = -\sec x$ and

$$g'(x) = 2\sec x,$$

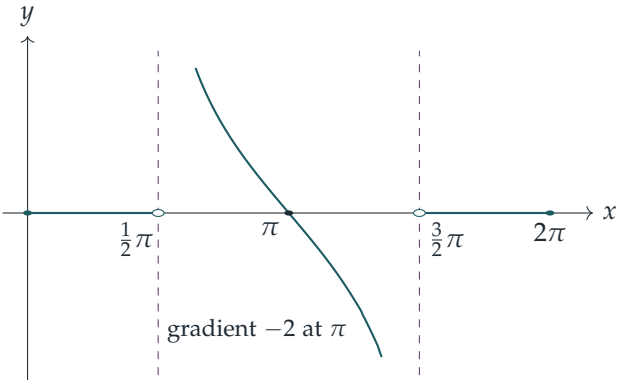
as required. On the other two intervals, $0 \le x < \frac{1}{2}\pi$ and $\frac{3}{2}\pi < x \le 2\pi$, we have $\cos x > 0$, so the same calculation gives $g'(x) = 0$.

The outer intervals. Here g is constant on each interval. Since $g(0) = \tanh^{-1} 0 - \sinh^{-1} 0 = 0$ and likewise $g(2\pi) = 0$, we get $g(x) = 0$ for $0 \le x < \frac{1}{2}\pi$ and for $\frac{3}{2}\pi < x \le 2\pi$.

The middle interval. Here $g'(x) = 2\sec x < 0$, so g is strictly decreasing. At the midpoint, $g(\pi) = \tanh^{-1} 0 - \sinh^{-1} 0 = 0$, and $g'(\pi) = 2\sec\pi = -2$. At the ends:

- as $x \to \frac{1}{2}\pi^+$, $\sin x \to 1$ so $\tanh^{-1}(\sin x) \to +\infty$, while $\tan x \to -\infty$ so $-\sinh^{-1}(\tan x) \to +\infty$; hence $g(x) \to +\infty$;
- as $x \to \frac{3}{2}\pi^-$, $\sin x \to -1$ and $\tan x \to +\infty$, so both terms send $g(x) \to -\infty$.

Also $g''(x) = 2\sec x \tan x$. For $\frac{1}{2}\pi < x < \pi$ both $\sec x$ and $\tan x$ are negative, so $g'' > 0$; for $\pi < x < \frac{3}{2}\pi$, $\sec x < 0 < \tan x$, so $g'' < 0$. So $(\pi, 0)$ is a point of inflection with gradient -2, and the curve has rotational symmetry about it (indeed $g'(2\pi - x) = g'(x)$ and $g(\pi) = 0$). In fact, since g and $2\tanh^{-1}(\sin x)$ have the same derivative on this interval and both vanish at π, $g(x) = 2\tanh^{-1}(\sin x) = 2\ln|\sec x + \tan x|$ there.



Checks

- With $\alpha = 0.6$: $f_\alpha(0) = \tan^{-1}(\cot 0.6) \approx 0.9708 = \frac{1}{2}\pi - 0.6$, and $f_\alpha(\pm 10^9) \approx -0.6$, as the asymptote predicts.
- $g(1) = \tanh^{-1}(\sin 1) - \sinh^{-1}(\tan 1) = 1.2261 - 1.2261 = 0$, confirming the flat pieces. A computer check of thousands of random points matched all the formulas above to within 10^{-6}.

Common slips

- Writing $f_\alpha(x) = \tan^{-1} x + c$ with one constant everywhere. Zero derivative only gives a constant on an interval.
- Writing $\sqrt{\sec^2 x} = \sec x$. It is $|\sec x|$, and that sign is the whole reason g is zero on two intervals and not on the third.

Beyond STEP

That zero derivative on an interval forces a constant is a consequence of the Mean Value Theorem, and it fails on sets that are not intervals, as both parts show. Part (i) is the addition formula for $\tan^{-1}$ with its correction by $\pm\pi$, the kind of branch bookkeeping that is central in complex analysis. Abbott's *Understanding Analysis* gives a careful account of the Mean Value Theorem and its consequences.

WORKED SOLUTION

STEP III 2021, Question 7

Pure • Complex Numbers in Geometry

STEP III 2021, Question 7

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

The key idea

When you add or subtract two points on the unit circle, $e^{i\theta} \pm e^{i\phi}$, take out the "average" factor $e^{i(\theta+\phi)/2}$. What is left is a pure cosine or a pure sine, which is what makes part (i) work. Part (i) then says that $(a+b)/(a-b)$ is purely imaginary, and in the complex plane "purely imaginary ratio" means "perpendicular". Everything after that is bookkeeping with sums of a, b, c, d.

Solution

Part (i)

Let $\alpha = \frac{1}{2}(\theta+\phi)$ and $\beta = \frac{1}{2}(\theta-\phi)$, so that $\theta = \alpha+\beta$ and $\phi = \alpha-\beta$. Then

$$e^{i\theta} + e^{i\phi} = e^{i\alpha}\left(e^{i\beta} + e^{-i\beta}\right) = 2\cos\beta\, e^{i\alpha},$$
$$e^{i\theta} - e^{i\phi} = e^{i\alpha}\left(e^{i\beta} - e^{-i\beta}\right) = 2i\sin\beta\, e^{i\alpha}.$$

The condition $\theta - \phi \neq 2n\pi$ says exactly that β is not a multiple of π, so $\sin\beta \neq 0$ and the denominator is non-zero. Dividing,

$$z = \frac{2\cos\beta}{2i\sin\beta} = -i\cot\beta = -i\cot\left(\tfrac{1}{2}(\theta-\phi)\right) = i\cot\left(\tfrac{1}{2}(\phi-\theta)\right),$$

using $1/i = -i$ and the fact that cot is odd. This is the required result.

Since $\cot\left(\frac{1}{2}(\phi-\theta)\right)$ is real, z lies on the imaginary axis:

$$|z| = \left|\cot\left(\tfrac{1}{2}(\phi-\theta)\right)\right|, \qquad \arg z = \begin{cases} \frac{1}{2}\pi & \text{if } \cot\left(\frac{1}{2}(\phi-\theta)\right) > 0, \\ -\frac{1}{2}\pi & \text{if } \cot\left(\frac{1}{2}(\phi-\theta)\right) < 0. \end{cases}$$

If $\cot\left(\frac{1}{2}(\phi-\theta)\right) = 0$, that is $\phi-\theta$ is an odd multiple of π, then $z = 0$ and its argument is undefined. (This happens when the two points are diametrically opposite, so their sum is zero.)

Part (ii)

Since A and B are on the unit circle centred at the origin, write $a = \mathrm{e}^{\mathrm{i}\theta}$ and $b = \mathrm{e}^{\mathrm{i}\phi}$. They are distinct, so $\theta - \phi$ is not a multiple of 2π, and part (i) applies:

$$\frac{x}{a-b} = \frac{a+b}{a-b} = \mathrm{i}k, \qquad k = \cot\Big(\tfrac{1}{2}(\phi-\theta)\Big) \in \mathbb{R}.$$

Because $x = a + b \neq 0$, we have $k \neq 0$. So $x = \mathrm{i}k\,(a-b)$: the vector $\overrightarrow{OX}$ (represented by x) is the vector $\overrightarrow{BA}$ (represented by $a-b$) rotated through a right angle and scaled by the non-zero real number k. Equivalently, $\arg x - \arg(a-b) = \pm\frac{1}{2}\pi$. Hence OX is perpendicular to AB.

Ask yourself Why do we need $a + b \neq 0$?

If $a + b = 0$ then X is the origin itself. There is no line OX, so there is nothing to be perpendicular. In the algebra this is the case $k = 0$, where the argument of x is undefined.

Part (iii)

With $h = a + b + c$ we have

$$h - a = b + c.$$

The complex number $h - a$ represents the vector $\overrightarrow{AH}$. There are two cases.

- If $b + c = 0$, then $h - a = 0$, so $h = a$.
- If $b + c \neq 0$, apply part (ii) to the distinct points B and C on the unit circle. It says that the vector represented by $b + c$ is perpendicular to BC. That vector is $\overrightarrow{AH}$, so AH is perpendicular to BC. (Here $H \neq A$, so AH is a genuine line.)

So either $h = a$ or $AH \perp BC$, as required. By symmetry the same holds for BH and CA, and for CH and AB, so H does lie on the altitudes. The case $h = a$ is when BC is a diameter: the triangle has a right angle at A, and its orthocentre is the vertex A.

Part (iv)

By the definition of the orthocentre,

$$p = a + b + c, \qquad q = b + c + d, \qquad r = c + d + a, \qquad s = d + a + b.$$

The midpoint of AQ is

$$\tfrac{1}{2}(a + q) = \tfrac{1}{2}(a + b + c + d).$$

This expression is symmetric in a, b, c, d, so it is also the midpoint of BR, of CS and of DP:

$$\tfrac{1}{2}(b + r) = \tfrac{1}{2}(c + s) = \tfrac{1}{2}(d + p) = \tfrac{1}{2}(a + b + c + d).$$

Let M be the point represented by $m = \frac{1}{2}(a + b + c + d)$. Then

$$q = 2m - a, \qquad r = 2m - b, \qquad s = 2m - c, \qquad p = 2m - d.$$

So the single map

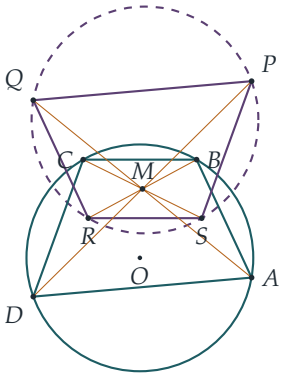
$$T : \ w \ \longmapsto \ 2m - w$$

sends $A \mapsto Q$, $B \mapsto R$, $C \mapsto S$ and $D \mapsto P$. It therefore maps the quadrilateral $ABCD$ onto $QRSP$, vertex by vertex.

To describe T, write it as $T(w) - m = -(w - m)$. The displacement from M is reversed, which is multiplication by $e^{i\pi}$ about M. So

T is a rotation through 180° *(a half-turn) about the point M with complex number* $\frac{1}{2}(a + b + c + d)$.

Equivalently, T is reflection in the point M. In particular $QRSP$ is congruent to $ABCD$ and, since a rotation preserves orientation, Q, R, S, P also run anticlockwise. The four orthocentres lie on the unit circle centred at the point with complex number $a + b + c + d$, the image of the circle through A, B, C, D.



Checks

- Part (i): with $\theta = 0$, $\phi = \frac{1}{2}\pi$, we get $z = (1 + \mathrm{i})/(1 - \mathrm{i}) = \mathrm{i}$, and $\mathrm{i}\cot\frac{1}{4}\pi = \mathrm{i}$. So $|z| = 1$ and $\arg z = \frac{1}{2}\pi$.
- Part (iv): take $a = 1$, $b = \mathrm{i}$, $c = -1$, $d = \frac{3}{5} - \frac{4}{5}\mathrm{i}$. Then $m = 0.3 + 0.1\mathrm{i}$, and $q = b + c + d = -0.4 + 0.2\mathrm{i}$, which equals $2m - a$ as claimed. For a square $(a, b, c, d = 1, \mathrm{i}, -1, -\mathrm{i})$ we get $m = 0$, and Q, R, S, P are just C, D, A, B: the half-turn is about O.
- All four parts were also confirmed by computer on thousands of random points on the circle.

Common slips

- Giving $\arg z = \frac{1}{2}\pi$ only. The sign of the cotangent decides between $\pm\frac{1}{2}\pi$, and $z = 0$ has no argument.
- In part (iv), matching the vertices in the wrong order. A goes to Q, not to P: the orthocentre opposite each vertex is the one of the triangle that *omits* it.

- Describing the map as "a rotation" without giving the centre and the angle, or as "an enlargement of scale factor -1" without giving the centre. Either description is fine, but it needs both pieces of information.

Beyond STEP

The neat formula $h = a + b + c$ (with the circumcentre at the origin) is the start of a whole style of complex-number geometry. For instance, the centroid is $\frac{1}{3}(a + b + c)$, so O, the centroid and H lie on a line, the Euler line. Part (iv) is a small piece of the same story for cyclic quadrilaterals. Tristan Needham's *Visual Complex Analysis* treats complex multiplication as rotation and scaling throughout, and is a superb next step.

WORKED SOLUTION

STEP III 2021, Question 8

Pure • Recurrence Relations and Limits

STEP III 2021, Question 8

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 2.8, *Squaring and Subtracting Two*: https://beyondstep.keithmcnulty.org/ch02/squaring-minus-two.html

The key idea

Everything rests on factorising the recurrence. Subtracting 2 gives $x_{n+1} - 2 = (x_n - 2)(x_n + 2)$, so once the terms are above 2 their distance from 2 is multiplied by more than 4 at every step. That gives part (i), and part (ii) follows. For part (iii), subtracting 4 from x_{n+1}^2 factorises just as neatly, and that is exactly what makes the product $x_1 x_2 \cdots x_n$ telescope.

Solution

Part (i)

We use induction on n. For $n = 1$ the claim reads $x_1 \geq 2 + 4^0(a-2) = a$, which is true with equality (note $4^0 = 1$).

Suppose that $x_n \geq 2 + 4^{n-1}(a-2)$ for some $n \geq 1$. Since $a > 2$, the right-hand side is greater than 2, so $x_n > 2$. Now

$$x_{n+1} - 2 = x_n^2 - 4 = (x_n - 2)(x_n + 2).$$

Here $x_n - 2 > 0$ and $x_n + 2 > 4$, so $(x_n - 2)(x_n + 2) \geq 4(x_n - 2)$. Using the inductive hypothesis $x_n - 2 \geq 4^{n-1}(a-2)$,

$$x_{n+1} - 2 \geq 4(x_n - 2) \geq 4 \cdot 4^{n-1}(a-2) = 4^n(a-2),$$

which is the claim for $n+1$. By induction, $x_n \geq 2 + 4^{n-1}(a-2)$ for all $n \geq 1$.

We never squared an inequality. Squaring $x_n \geq (\text{bound})$ is only valid when both sides are non-negative, and the factorised form avoids that worry entirely.

Part (ii)

If $|a| > 2$. Then $x_2 = a^2 - 2 > 4 - 2 = 2$. The sequence $x_2, x_3, \ldots$ obeys the same recurrence and starts above 2, so part (i), applied with x_2 in place of a (shifting the index by one), gives

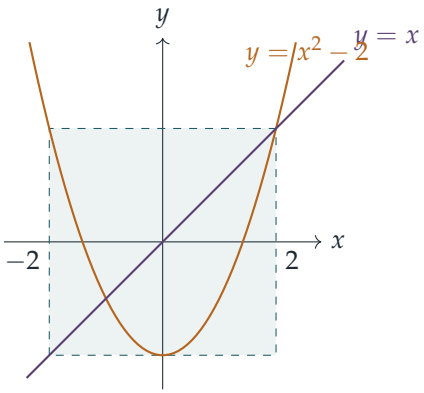
$$x_n \geq 2 + 4^{n-2}(x_2 - 2) \qquad (n \geq 2).$$

Since $x_2 - 2 > 0$ and $4^{n-2} \to \infty$, we get $x_n \to \infty$.

If $|a| \leq 2$. We show by induction that $-2 \leq x_n \leq 2$ for every n. This holds for $n = 1$. If $-2 \leq x_n \leq 2$ then $0 \leq x_n^2 \leq 4$, so $-2 \leq x_n^2 - 2 \leq 2$, which means $-2 \leq x_{n+1} \leq 2$. So the sequence is bounded and cannot tend to infinity.

Ask yourself In the case $|a| \leq 2$, is it enough to show that $x_n \leq 2$ for all n?

No. An upper bound alone is not enough, because a term below -2 would square to something larger than 4, and the following term would be above 2. That is why the induction has to carry both bounds together. The graph shows why it works: the parabola $y = x^2 - 2$ maps the interval $[-2, 2]$ onto itself.



Putting the two cases together, $x_n \to \infty$ if and only if $|a| > 2$.

Part (iii)

The factorisation that drives everything here is

$$x_{n+1}^2 - 4 = (x_n^2 - 2)^2 - 4 = x_n^4 - 4x_n^2 = x_n^2\,(x_n^2 - 4).$$

Since $a > 2$, part (i) gives $x_n > 2$ for all n, so $x_n > 0$ and $x_n^2 - 4 > 0$. Taking positive square roots,

$$\sqrt{x_{n+1}^2 - 4} = x_n \sqrt{x_n^2 - 4}.$$

Applying this repeatedly (formally, by induction on n),

$$\sqrt{x_{n+1}^2 - 4} = x_n \sqrt{x_n^2 - 4} = x_n x_{n-1} \sqrt{x_{n-1}^2 - 4} = \cdots = x_1 x_2 \cdots x_n \sqrt{a^2 - 4}.$$

Therefore, for every $n \geq 1$,

$$\frac{\sqrt{x_{n+1}^2 - 4}}{x_{n+1}} = \frac{\sqrt{a^2-4}\, x_1 x_2 \cdots x_n}{x_{n+1}},$$

and comparing with the definition of y_n shows that the required formula holds for all n exactly when $A = \sqrt{a^2-4}$.

Finding a. The value $A = \sqrt{a^2-4}$ is forced. At $n = 1$ the required formula reads $Aa/x_2 = a\sqrt{a^2-4}/x_2$, and since $a > 0$ and $x_2 > 0$ this gives $A = \sqrt{a^2-4}$. Conversely, $A = \sqrt{a^2-4}$ with $a > 2$ means $A^2 = a^2 - 4$, so

$$a = \sqrt{A^2+4}.$$

This value does satisfy $a > 2$, because $A > 0$ gives $A^2 + 4 > 4$. With this a we have $\sqrt{a^2-4} = A$, and the displayed identity above becomes $y_n = \sqrt{x_{n+1}^2 - 4}\,/\,x_{n+1}$ for all $n \geq 1$, as required.

Convergence. With this value of a, since $x_{n+1} > 0$,

$$y_n = \frac{\sqrt{x_{n+1}^2 - 4}}{x_{n+1}} = \sqrt{1 - \frac{4}{x_{n+1}^2}}.$$

As $a > 2$, part (ii) gives $x_{n+1} \to \infty$, so $4/x_{n+1}^2 \to 0$ and $y_n \to 1$. The second sequence converges, and its limit is 1.

Checks

- Take $A = 1$, so $a = \sqrt{5} \approx 2.236$. Then $x_2 = 3$, $x_3 = 7$ and $y_1 = \sqrt{5}/3 \approx 0.745$. From the formula, $\sqrt{9-4}/3 = \sqrt{5}/3$, which agrees. Next, $y_2 = \sqrt{5} \cdot 3/7 \approx 0.958$ and $\sqrt{49-4}/7 = 3\sqrt{5}/7$, which also agrees. The terms $y_3 \approx 0.99909$ and $y_4 \approx 0.9999996$ are racing towards 1.
- The substitution $a = t + 1/t$ with $t > 1$ gives $x_n = t^{2^{n-1}} + t^{-2^{n-1}}$. This shows how fast the growth really is (doubly exponential, far faster than the 4^{n-1} of part (i)), and it is consistent with every result above.
- The bound in part (i) and the boundedness in part (ii) were also tested by computer on many random starting values.

Common slips

- Squaring the inequality from the inductive hypothesis without first checking that both sides are non-negative.
- In part (ii), bounding only $x_n \leq 2$ when $|a| \leq 2$. You also need $x_n \geq -2$, or the next term could jump above 2.
- Working backwards to find a but never confirming that this value works for all n and satisfies $a > 2$.

Beyond STEP

The map $x \mapsto x^2 - 2$ is the most famous member of the quadratic family $x \mapsto x^2 + c$. On $[-2, 2]$, putting $x = 2\cos\theta$ turns it into angle-doubling, $\theta \mapsto 2\theta$, a standard first example of chaotic dynamics. Outside that interval every orbit escapes to infinity. For $a > 2$ the same idea works with $x = 2\cosh u$, and then the product in part (iii) is a hyperbolic version of the classical $\prod \cos(\theta/2^k)$ telescoping. For the analysis behind the limits and inductions here, read Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP III 2021, Question 9

Mechanics • Oblique Impacts with Restitution

STEP III 2021, Question 9

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

The key idea

The geometry is two applications of the sine rule, in the corner triangles CPQ and BRP; the shared height of QR above CB cancels when we multiply. For the bounces, a smooth wall keeps the velocity component along it and multiplies the component across it by e, so the angles the path makes with the wall before and after satisfy $\tan(\text{after}) = e\tan(\text{before})$. Each bounce then gives one of the two brackets in the first result.

Solution

The geometric identity

Let h be the distance of the line QR from BC. Because $QR \parallel CB$, triangle AQR is equilateral, so $CQ = BR = h/\sin 60^\circ = 2h/\sqrt{3}$.

In triangle CPQ the angle at C is 60° and the angle at P is θ, so the angle at Q is $120^\circ - \theta$. The sine rule gives

$$\frac{CP}{\sin(120^\circ - \theta)} = \frac{CQ}{\sin\theta} \quad \Longrightarrow \quad (a - x)\sin\theta = \frac{2h}{\sqrt{3}}\left(\tfrac{\sqrt{3}}{2}\cos\theta + \tfrac{1}{2}\sin\theta\right).$$

Dividing by $\sin\theta$ (non-zero, as $0 < \theta < 180^\circ$) and tidying,

$$\sqrt{3}\cot\theta + 1 = \frac{\sqrt{3}\,(a - x)}{h}.$$

In triangle BRP the angle at B is 60°, the angle at R is ϕ and the angle at P is $120^\circ - \phi$. The sine rule gives

$$\frac{BP}{\sin\phi} = \frac{BR}{\sin(120^\circ - \phi)} \quad \Longrightarrow \quad x\left(\tfrac{\sqrt{3}}{2}\cos\phi + \tfrac{1}{2}\sin\phi\right) = \frac{2h}{\sqrt{3}}\sin\phi,$$

so that

$$\sqrt{3}\cot\phi + 1 = \frac{4h}{\sqrt{3}\,x}.$$

Multiplying the two results, h and $\sqrt{3}$ cancel:

$$(\sqrt{3}\cot\phi + 1)(\sqrt{3}\cot\theta + 1) = \frac{4(a-x)}{x},$$

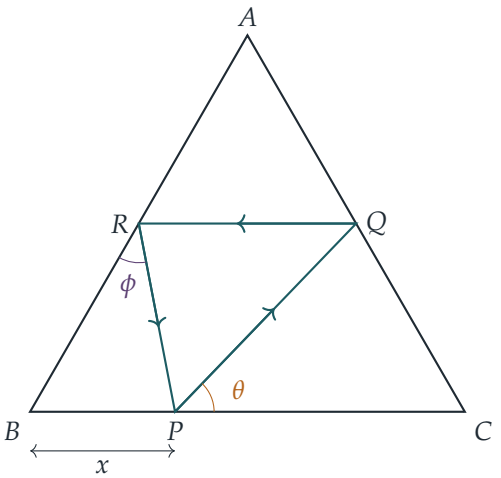
which is the required identity once we multiply by x.

The bounces

Model the ball as a particle moving on the smooth table, so it travels in straight lines between impacts, and the frame as smooth, so each impact gives only an impulse perpendicular to the wall. Suppose the ball meets a wall with speed u at angle α to it, and leaves with speed v at angle β to it. The component along the wall is unchanged and the component across it is reversed and multiplied by e:

$$v\cos\beta = u\cos\alpha, \qquad v\sin\beta = eu\sin\alpha \quad\Longrightarrow\quad \tan\beta = e\tan\alpha.$$

The ball also keeps moving the same way along the wall, so both α and β are acute when measured from that direction. (We need $e > 0$, otherwise the ball would slide along the wall after the first impact.)



At Q. The ball arrives along PQ, at angle $\angle PQC = 120^\circ - \theta$ to the wall CA, moving towards A. It leaves along QR, at angle $\angle AQR = 60^\circ$ to the wall (corresponding angles, as $QR \parallel CB$). So

$$\tan 60^\circ = e\tan(120^\circ - \theta).$$

Using the subtraction formula with $\tan 120^\circ = -\sqrt{3}$ and then multiplying top and bottom by $\cot\theta$,

$$\tan(120^\circ - \theta) = \frac{-\sqrt{3} - \tan\theta}{1 - \sqrt{3}\tan\theta} = \frac{\sqrt{3}\cot\theta + 1}{\sqrt{3} - \cot\theta}.$$

Write $c = \sqrt{3}\cot\theta$. The bounce condition $\sqrt{3}(\sqrt{3} - \cot\theta) = e(\sqrt{3}\cot\theta + 1)$ becomes $3 - c = e(c+1)$, that is, $4 - (c+1) = e(c+1)$, so

$$\sqrt{3}\cot\theta + 1 = \frac{4}{1+e}.$$

At R. The ball arrives along QR, at angle $\angle ARQ = 60^\circ$ to the wall AB, moving towards B, and leaves along RP at angle $\angle BRP = \phi$. So

$$\tan\phi = e\tan 60^\circ = \sqrt{3}\,e, \qquad \text{so} \qquad \sqrt{3}\cot\phi = \frac{1}{e}, \qquad \sqrt{3}\cot\phi + 1 = \frac{1+e}{e}.$$

Finding x. Substituting both brackets into the identity,

$$\frac{1+e}{e} \cdot \frac{4}{1+e} \cdot x = 4(a-x) \quad \Longrightarrow \quad x = e(a-x) \quad \Longrightarrow \quad x = \frac{ae}{1+e},$$

as required.

Going round again

Ask yourself What has to happen at P for the ball to set off along PQ a second time?

The ball reaches P along RP. In triangle BRP this path makes angle $\angle RPB = 120^\circ - \phi$ with BC, so it arrives moving towards C only if this angle is acute. To retrace PQ, it must leave at angle θ to PC, so we need

$$\tan\theta = e\tan(120^\circ - \phi).$$

From the work at Q, $\sqrt{3}\cot\theta = \frac{4}{1+e} - 1 = \frac{3-e}{1+e}$, so $\tan\theta = \dfrac{\sqrt{3}(1+e)}{3-e}$, which is positive: PQ does head towards the C side.

The same formula as at Q, with ϕ in place of θ and $\sqrt{3}\cot\phi = \frac{1}{e}$, gives

$$\tan(120^\circ - \phi) = \frac{\sqrt{3}\,(\sqrt{3}\cot\phi + 1)}{3 - \sqrt{3}\cot\phi} = \frac{\sqrt{3}\,(1+e)/e}{(3e-1)/e} = \frac{\sqrt{3}(1+e)}{3e-1}.$$

If $e \le \frac{1}{3}$, then $120^\circ - \phi \ge 90^\circ$, so after the impact the ball moves straight away from BC or back towards B. It cannot then follow PQ, which leans towards C. So $e > \frac{1}{3}$, and the condition becomes

$$\frac{\sqrt{3}(1+e)}{3-e} = \frac{\sqrt{3}\,e(1+e)}{3e-1} \quad \Longleftrightarrow \quad 3e - 1 = 3e - e^2 \quad \Longleftrightarrow \quad e^2 = 1.$$

Since $e > 0$, this gives $e = 1$, as required. The ball's speed changes at each impact (unless $e = 1$), but speed plays no part here: only the directions matter.

Checks

- When $e = 1$: $x = \frac{a}{2}$ and $\tan\theta = \tan\phi = \sqrt{3}$, so $\theta = \phi = 60^\circ$ and PQR is the triangle joining the midpoints of the sides. By symmetry it is a closed perfectly elastic orbit, as expected.
- With $a = 1$ and $e = \frac{1}{2}$: $x = \frac{1}{3}$ and $\theta \approx 46.10^\circ$. Stepping the ball through the three impacts by computer brings it back to P exactly. After the next impact it leaves at about 68.95° to BC, not 46.10°, so it does not go round PQR again.
- As $e \to 0$, $x \to 0$: P slides into the corner B. This fits the ball losing almost all its velocity across each wall.

Common slips

- Applying e to the whole velocity, or to the component along the wall. Only the component perpendicular to the smooth wall changes.
- Measuring the angle of incidence from the normal and then using $\tan\beta = e\tan\alpha$, which holds for angles measured from the wall. Measured from the normal, the law is $\cot\beta = e\cot\alpha$.
- In the final part, forgetting that $120^\circ - \phi$ might be obtuse, or that $e = -1$ must be rejected.

Beyond STEP

Paths of a ball bouncing inside a region are the subject of *mathematical billiards*. With $e = 1$ the triangle joining the midpoints is the Fagnano orbit. In any acute triangle this orbit joins the feet of the altitudes, and it is the inscribed triangle of shortest perimeter. With $e < 1$ each bounce flattens the path towards the wall, and as we found here, the ball cannot repeat a closed circuit. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats impulse and collisions carefully. For billiards themselves, Tabachnikov's *Geometry and Billiards* is a lovely next step.

WORKED SOLUTION

STEP III 2021, Question 10

Mechanics • Wrapping String Motion

STEP III 2021, Question 10

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

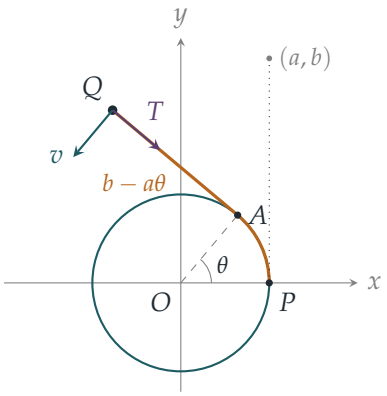
The key idea

Describe everything by the angle θ through which the straight part of the string has turned. The string leaves the cylinder along a tangent, so the particle sits at the point of tangency plus a tangent vector of length $b - a\theta$. Differentiating shows that the particle's velocity is always at right angles to the string, so the tension does no work and the speed never changes. At each instant the particle is moving on a circle centred at the point where the string leaves the cylinder, and the tension supplies the centripetal force.

Solution

We model the particle as a point moving on a smooth horizontal table, so the only horizontal force on it is the tension T in the light inextensible string (the weight is balanced by the normal reaction of the table). The string is taut throughout.

For the string to wrap round the cylinder, the impulse must send the particle in the negative x-direction from (a, b), so the string turns anticlockwise. Let A be the point where the straight part of the string leaves the cylinder at time t.



Part (i)

The point A is $(a\cos\theta, a\sin\theta)$, since the radius to P has turned through θ. The length of string wrapped on the cylinder is the arc PA, of length $a\theta$, so the straight part has length $b - a\theta$. It lies along the tangent at A, in the direction of increasing θ, which is the unit vector $(-\sin\theta, \cos\theta)$. (At $\theta = 0$ this is $(0, 1)$, matching the initial position (a, b).) So the particle is at

$$x = a\cos\theta - (b - a\theta)\sin\theta, \qquad y = a\sin\theta + (b - a\theta)\cos\theta.$$

Differentiating with respect to t, using the product rule and $\frac{\mathrm{d}}{\mathrm{d}t}(b - a\theta) = -a\dot{\theta}$,

$$\dot{x} = -a\dot{\theta}\sin\theta + a\dot{\theta}\sin\theta - (b - a\theta)\dot{\theta}\cos\theta = -(b - a\theta)\dot{\theta}\cos\theta,$$
$$\dot{y} = a\dot{\theta}\cos\theta - a\dot{\theta}\cos\theta - (b - a\theta)\dot{\theta}\sin\theta = -(b - a\theta)\dot{\theta}\sin\theta.$$

Hence $\dot{x}^2 + \dot{y}^2 = (b - a\theta)^2\dot{\theta}^2$, and since $b - a\theta > 0$ and $\dot{\theta} > 0$ (the string is wrapping on), the speed is

$$v = (b - a\theta)\dot{\theta}.$$

Part (ii)

Ask yourself Why should the speed stay equal to u? The string is pulling on the particle the whole time.

From part (i) the velocity is $-(b - a\theta)\dot{\theta}\,(\cos\theta, \sin\theta)$, which is parallel to the radius OA and therefore perpendicular to the string, whose direction is $(-\sin\theta, \cos\theta)$. The tension acts along the string, so its rate of doing work, $\mathbf{T}\cdot\mathbf{v}$, is zero at every instant. No other horizontal force acts, so the kinetic energy, and hence the speed, is constant. Initially the speed is u, so

$$\dot{\theta}\,(b - a\theta) = u,$$

as required.

Part (iii)

Using (ii), the velocity is $\mathbf{v} = -u(\cos\theta, \sin\theta)$ with u constant, so the acceleration is

$$\dot{\mathbf{v}} = -u\dot{\theta}(-\sin\theta, \cos\theta).$$

This has magnitude $u\dot{\theta}$ and points from the particle towards A, exactly along the string. (Equivalently: at each instant the particle is moving on a circle of radius $b - a\theta$ about A at speed u, so its acceleration is $u^2/(b - a\theta)$ towards A.) Newton's second law along the string gives

$$T = mu\dot{\theta} = \frac{mu^2}{b - a\theta}.$$

It remains to express $b - a\theta$ in terms of t. Integrating (ii) with respect to t, using $\theta = 0$ at $t = 0$,

$$b\theta - \tfrac{1}{2}a\theta^2 = ut.$$

Then

$$(b - a\theta)^2 = b^2 - 2a\left(b\theta - \tfrac{1}{2}a\theta^2\right) = b^2 - 2aut,$$

and since $b - a\theta > 0$ we take the positive root, $b - a\theta = \sqrt{b^2 - 2aut}$. Therefore

$$T = \frac{mu^2}{\sqrt{b^2 - 2aut}},$$

as required.

Checks

- At $t = 0$ the tension is mu^2/b: the particle starts moving on a circle of radius b about P at speed u, as expected.
- The string is completely wound when $a\theta = b$, which happens at $t = b^2/(2au)$; then $b^2 - 2aut = 0$ and the tension grows without bound, as the radius of the instantaneous circle shrinks to zero at constant speed.
- Dimensions: b^2 and aut are both areas, and mu^2/length is a force.
- We confirmed the results by computer, stepping the motion numerically for several random choices of a, b, u and m and differentiating the position numerically: the speed stayed at u and the measured force along the string matched $mu^2/\sqrt{b^2 - 2aut}$ to about 3×10^{-5} (relative).

Common slips

- Assuming angular momentum about O is conserved. It isn't: the tension acts along a tangent to the cylinder, which does not pass through O, so it has a moment about O. Energy is the conserved quantity here.
- Using the length b instead of $b - a\theta$ for the straight part, forgetting the string already wrapped on the cylinder.
- Taking the radius OQ of the particle's path, rather than the straight length AQ, as the radius of curvature in the tension calculation.

Beyond STEP

The path traced by the end of a string unwinding from (or winding onto) a curve is called an *involute*; the involute of a circle is the shape of most gear-tooth profiles, because it makes meshing gears transmit motion smoothly. The question also illustrates a general principle met in first-year mechanics: a constraint force perpendicular to the velocity does no work, so energy can be conserved even when momentum and angular momentum are not. Kleppner and Kolenkow, *An Introduction to Mechanics*, has a good discussion of exactly this kind of problem.

WORKED SOLUTION

STEP III 2021, Question 11

Probability and Statistics • Exponential Distribution

STEP III 2021, Question 11

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 12.18, *Memorylessness*: https://beyondstep.keithmcnulty.org/ch12/memoryless.html

The key idea

Every question about Y and Z is really a question about which interval X lands in. The event $Y = n$ is $n \leq X < n+1$, and the event $Z < z$ splits into the disjoint pieces $n \leq X < n+z$, one for each n. The exponential distribution's tail $\mathrm{P}(X \geq x) = \mathrm{e}^{-\lambda x}$ turns each piece into a term of a geometric series.

Solution

Throughout we write $q = \mathrm{e}^{-\lambda}$, so $0 < q < 1$ because $\lambda > 0$. For $x \geq 0$,

$$\mathrm{P}(X \geq x) = \int_x^\infty \lambda \mathrm{e}^{-\lambda t}\,\mathrm{d}t = \mathrm{e}^{-\lambda x},$$

and since X is continuous, $\mathrm{P}(a \leq X < b) = \mathrm{P}(a < X < b) = \mathrm{e}^{-\lambda a} - \mathrm{e}^{-\lambda b}$ for $0 \leq a < b$.

Part (i)

Y is the integer part of X, so $Y = n$ exactly when $n \leq X < n+1$. Hence

$$\mathrm{P}(Y = n) = \mathrm{e}^{-n\lambda} - \mathrm{e}^{-(n+1)\lambda} = (1 - \mathrm{e}^{-\lambda})\mathrm{e}^{-n\lambda},$$

as required. (So Y is geometric, starting at 0, with ratio q.)

Part (ii)

$Z = X - Y$ is the fractional part of X, so $0 \leq Z < 1$. Take $0 \leq z \leq 1$. The event $Z < z$ happens exactly when, for the integer $n = Y$, we have $n \leq X < n + z$. These events for

$n = 0, 1, 2, \ldots$ are disjoint, so

$$\begin{aligned} \mathrm{P}(Z < z) &= \sum_{n=0}^{\infty} \mathrm{P}(n \le X < n + z) = \sum_{n=0}^{\infty} \left(\mathrm{e}^{-n\lambda} - \mathrm{e}^{-n\lambda - \lambda z}\right) \\ &= (1 - \mathrm{e}^{-\lambda z}) \sum_{n=0}^{\infty} q^n = \frac{1 - \mathrm{e}^{-\lambda z}}{1 - \mathrm{e}^{-\lambda}}, \end{aligned}$$

summing the geometric series, which converges because $0 < q < 1$. This is the required result.

Part (iii)

Differentiating the distribution function from (ii), Z has density

$$g(z) = \frac{\lambda \mathrm{e}^{-\lambda z}}{1 - \mathrm{e}^{-\lambda}} \qquad (0 \le z \le 1),$$

and $g(z) = 0$ otherwise. Integrating by parts with $u = z$ and $\mathrm{d}v = \lambda \mathrm{e}^{-\lambda z}\,\mathrm{d}z$,

$$\int_0^1 z\lambda \mathrm{e}^{-\lambda z}\,\mathrm{d}z = \left[-z\mathrm{e}^{-\lambda z}\right]_0^1 + \int_0^1 \mathrm{e}^{-\lambda z}\,\mathrm{d}z = -\mathrm{e}^{-\lambda} + \frac{1 - \mathrm{e}^{-\lambda}}{\lambda}.$$

Dividing by $1 - \mathrm{e}^{-\lambda}$,

$$\mathrm{E}(Z) = \frac{1}{\lambda} - \frac{\mathrm{e}^{-\lambda}}{1 - \mathrm{e}^{-\lambda}} = \frac{1}{\lambda} - \frac{1}{\mathrm{e}^{\lambda} - 1}.$$

Part (iv)

Let $0 \le z_1 < z_2 \le 1$. The event $\{Y = n \text{ and } z_1 < Z < z_2\}$ is exactly $n + z_1 < X < n + z_2$, so

$$\mathrm{P}(Y = n \text{ and } z_1 < Z < z_2) = \mathrm{e}^{-\lambda(n+z_1)} - \mathrm{e}^{-\lambda(n+z_2)} = \mathrm{e}^{-n\lambda}\left(\mathrm{e}^{-\lambda z_1} - \mathrm{e}^{-\lambda z_2}\right).$$

> **Ask yourself** What would we need to compare this with to decide about independence?

With the product $\mathrm{P}(Y = n)\,\mathrm{P}(z_1 < Z < z_2)$. Since Z is continuous, $\mathrm{P}(Z = z_1) = 0$, so by (ii)

$$\mathrm{P}(z_1 < Z < z_2) = \mathrm{P}(Z < z_2) - \mathrm{P}(Z < z_1) = \frac{\mathrm{e}^{-\lambda z_1} - \mathrm{e}^{-\lambda z_2}}{1 - \mathrm{e}^{-\lambda}}.$$

Multiplying by $\mathrm{P}(Y = n) = (1 - \mathrm{e}^{-\lambda})\mathrm{e}^{-n\lambda}$ from (i), the factors $1 - \mathrm{e}^{-\lambda}$ cancel and we get exactly the joint probability above. So

$$\mathrm{P}(Y = n \text{ and } z_1 < Z < z_2) = \mathrm{P}(Y = n)\,\mathrm{P}(z_1 < Z < z_2)$$

for every non-negative integer n and every interval $(z_1, z_2) \subseteq [0, 1]$. Since Y only takes non-negative integer values and Z only takes values in $[0, 1)$, events of these forms determine the whole joint distribution. Therefore Y and Z *are* independent: knowing which unit interval X fell in tells us nothing about where in that interval it fell.

Checks

- Putting $z = 1$ in (ii) gives $\mathrm{P}(Z < 1) = 1$, as it must, and $z = 0$ gives 0.
- As $\lambda \to 0$ the density of X is almost flat over each unit interval, so Z should be close to uniform on $[0, 1]$. Indeed $\mathrm{e}^{\lambda} - 1 = \lambda + \frac{1}{2}\lambda^2 + \cdots$ gives $\frac{1}{\mathrm{e}^{\lambda}-1} = \frac{1}{\lambda} - \frac{1}{2} + O(\lambda)$, so $\mathrm{E}(Z) \to \frac{1}{2}$. As $\lambda \to \infty$, X is almost always tiny and $\mathrm{E}(Z) \to 0$.
- For $\lambda = 1$, $\mathrm{E}(Z) = 1 - \frac{1}{\mathrm{e}-1} \approx 0.4180$, which matches a computer simulation of 200 000 values of X; the simulated joint frequencies in (iv) also matched the products of the marginals.

Common slips

- Writing $\mathrm{P}(Z < z) = \mathrm{P}(X < z)$, forgetting that $Z < z$ can happen in every unit interval, not just the first.
- Concluding independence from a single choice of n, z_1, z_2: the factorisation must hold for all of them.

Beyond STEP

The independence in (iv) is a shadow of the *memoryless property* of the exponential distribution: given $X \geq n$, the excess $X - n$ is again exponential with the same rate, whatever n is. This is why the exponential distribution models waiting times in Poisson processes, and why its integer part is geometric, the discrete memoryless distribution. Grimmett and Welsh, *Probability: An Introduction*, develops both distributions and the Poisson process.

WORKED SOLUTION

STEP III 2021, Question 12

Probability and Statistics • Variance of Sums

STEP III 2021, Question 12

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 12.4, *Independent in Pairs, Not All Together*: https://beyondstep.keithmcnulty.org/ch12/pairwise-independence.html

The key idea

Write the team's score as a sum of indicator-type variables, one for each pair of players, and expand the variance of the sum. Everything then rests on how two *different* pairs interact. If the pairs have no player in common they are plainly independent, so the real work is with pairs that share one player. In (i) such pairs turn out to be independent, so the cross terms vanish. In (iii) they are not, and the identity in (ii) is the tool for counting what they contribute.

Solution

Part (i)

Each die is fair and the players roll independently. Two given players roll the same number with probability $\frac{1}{6}$, whatever the first of them rolls. So

$$\mathrm{P}(X_{12} = 1) = \mathrm{P}(X_{23} = 1) = \tfrac{1}{6}.$$

Both $X_{12} = 1$ and $X_{23} = 1$ happen exactly when players 1, 2 and 3 all roll the same number, so

$$\mathrm{P}(X_{12} = 1,\ X_{23} = 1) = \tfrac{1}{6} \cdot \tfrac{1}{6} = \tfrac{1}{36} = \mathrm{P}(X_{12} = 1)\,\mathrm{P}(X_{23} = 1).$$

Subtracting this from the marginal probabilities gives the other three cells. For example, $\mathrm{P}(X_{12} = 1, X_{23} = 0) = \frac{1}{6} - \frac{1}{36} = \frac{1}{6} \cdot \frac{5}{6}$, and similarly $\mathrm{P}(X_{12} = 0, X_{23} = 1) = \frac{5}{6} \cdot \frac{1}{6}$ and $\mathrm{P}(X_{12} = 0, X_{23} = 0) = 1 - \frac{1}{6} - \frac{1}{6} + \frac{1}{36} = \frac{5}{6} \cdot \frac{5}{6}$. So the joint distribution is the product of the marginals, and X_{12} and X_{23} are independent.

Ask yourself Why does knowing $X_{12} = 1$ tell us nothing about X_{23}?

Given that players 1 and 2 match, player 2's number is still uniform on $\{1, \ldots, 6\}$, and player 3 matches it with probability $\frac{1}{6}$, exactly as before. The shared player carries no information because every face is equally likely.

Let $T = \sum_{i<j} X_{ij}$ be the total score, a sum of $N = \binom{n}{2}$ terms. Each X_{ij} is Bernoulli with parameter $\frac{1}{6}$, so $\mathrm{E}(X_{ij}) = \frac{1}{6}$ and $\mathrm{Var}(X_{ij}) = \frac{1}{6} \cdot \frac{5}{6} = \frac{5}{36}$. Hence

$$\mathrm{E}(T) = \frac{N}{6} = \frac{n(n-1)}{12}.$$

For the variance, take any two different pairs. If they have no player in common, they depend on disjoint sets of dice and so are independent. If they share one player, they are independent by the argument above, after relabelling the players. So the X_{ij} are pairwise independent, every covariance is zero, and the variance of the sum is the sum of the variances:

$$\mathrm{Var}(T) = N \cdot \frac{5}{36} = \frac{5n(n-1)}{72}.$$

Part (ii)

Let $S = Y_1 + \cdots + Y_m$. Then $\mathrm{E}(S) = 0$ by linearity, so $\mathrm{Var}(S) = \mathrm{E}(S^2) - 0^2 = \mathrm{E}(S^2)$. Expanding the square, each Y_i^2 appears once and each product $Y_i Y_j$ with $i < j$ appears twice:

$$S^2 = \sum_{i=1}^{m} Y_i^2 + 2 \sum_{i=1}^{m-1} \sum_{j=i+1}^{m} Y_i Y_j.$$

Taking expectations and using linearity gives the required formula.

Part (iii)

For two players, $\mathrm{P}(\text{same even number}) = \frac{3}{36} = \frac{1}{12}$, and likewise $\mathrm{P}(\text{same odd number}) = \frac{1}{12}$. So

$$\mathrm{P}(Z_{ij} = 1) = \mathrm{P}(Z_{ij} = -1) = \tfrac{1}{12}, \qquad \mathrm{P}(Z_{ij} = 0) = \tfrac{5}{6}.$$

Not independent. If $Z_{12} = 1$ and $Z_{23} = -1$, player 2's number would have to be both even and odd. So

$$\mathrm{P}(Z_{12} = 1,\ Z_{23} = -1) = 0 \neq \tfrac{1}{12} \cdot \tfrac{1}{12} = \mathrm{P}(Z_{12} = 1)\,\mathrm{P}(Z_{23} = -1),$$

and Z_{12} and Z_{23} are not independent.

Mean. By symmetry $\mathrm{E}(Z_{ij}) = \frac{1}{12} - \frac{1}{12} = 0$, so the total score $U = \sum_{i<j} Z_{ij}$ has mean 0.

Variance. Every Z_{ij} has mean zero, so we can use (ii) with the $N = \binom{n}{2}$ variables Z_{ij}. We need two kinds of expectation.

- Squares: Z_{ij}^2 is 1 exactly when players i and j match, so $\mathrm{E}(Z_{ij}^2) = \frac{1}{6}$.

- Products of two different pairs. If the pairs have no player in common, the two variables are independent with mean zero, so $\mathrm{E}(Z_{ij}Z_{kl}) = \mathrm{E}(Z_{ij})\,\mathrm{E}(Z_{kl}) = 0$. If they share a player, say $Z_{12}Z_{23}$, the product is nonzero only when all three players roll the same number. Then both factors are $+1$ (even) or both are -1 (odd), so the product is 1. Hence $\mathrm{E}(Z_{12}Z_{23}) = \frac{1}{36}$.

Ask yourself How many unordered pairs of pairs share exactly one player?

Choose the shared player in n ways, then choose its two partners from the remaining $n-1$ players in $\binom{n-1}{2}$ ways. This gives $n\binom{n-1}{2} = \frac{1}{2}n(n-1)(n-2)$ such pairs of pairs, each counted once, which is what the double sum in (ii) needs.

Putting this together,

$$\begin{aligned}\mathrm{Var}(U) &= \binom{n}{2}\cdot\frac{1}{6} + 2\cdot\frac{n(n-1)(n-2)}{2}\cdot\frac{1}{36}\\ &= \frac{n(n-1)}{36}\big(3+(n-2)\big)\\ &= \frac{n(n-1)(n+1)}{36} = \frac{1}{36}n(n^2-1),\end{aligned}$$

as required.

Checks

- For the example roll $3,3,3,3,6,6,2$, the pair count gives $6+1=7$ and the signed count gives $-6+1=-5$, matching the question.
- With $n=2$ there is only one pair, so $\mathrm{Var}(U) = \mathrm{E}(Z_{12}^2) = \frac{1}{6}$, and the formula gives $\frac{2\cdot 3}{36} = \frac{1}{6}$.
- For $n=3$, $U = \pm 3$ when all three dice agree (probability $\frac{1}{72}$ each). Otherwise exactly one pair matches and $U = \pm 1$, with total probability $\frac{15}{36}$. So $\mathrm{E}(U^2) = 9\cdot\frac{1}{36} + \frac{15}{36} = \frac{2}{3}$, agreeing with $\frac{3\cdot 8}{36}$. Exact enumeration by computer of all outcomes for $n \le 6$ confirms both means and both variances; for example $n=6$ gives $\frac{25}{12}$ in (i) and $\frac{35}{6}$ in (iii).

Common slips

- Claiming independence of X_{12} and X_{23} after checking only one cell, without saying why the other cells follow. For two-valued variables they do follow, but say so.
- In (iii), forgetting that pairs sharing a player contribute $+\frac{1}{36}$ whether the common number is even or odd. The minus signs cancel in the product.
- Miscounting the overlapping pairs of pairs, for example using $n(n-1)(n-2)$ (ordered) and then also multiplying by 2.

Beyond STEP

Part (i) is a classic example of variables that are *pairwise* independent but not *mutually* independent: if $X_{12} = X_{23} = 1$ then $X_{13} = 1$ is forced. Pairwise independence is all a variance calculation needs, an idea used in randomised algorithms and in the birthday problem. Part (iii) shows the general method of covariances of sums when the terms are dependent. Both are developed in Blitzstein and Hwang, *Introduction to Probability*.

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